The A-/Ā-distinction is categorial: The view from A-resumption in Iraqi Arabic[*]

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Abstract. This paper investigates the A-/Ā-distinction through the novel lens of base-generated resumptive A-dependencies in Iraqi Arabic. A-resumption presents a challenge to traditional positional approaches to the A-/Ā-distinction that distinguish just two types of positions (i.e. A vs. Ā) because A-resumption displays a mixture of A- and Ā-properties. Moreover, A-resumption presents a challenge to approaches which derive A- and Ā-properties from the mechanics of movement because A-resumption does not involve movement. Instead, I develop a featural analysis of A-resumption and the A-/Ā-distinction according to which (i) differences between A- and Ā-dependencies are linked to the category of the head of the dependency—D and Q, respectively—and (ii) differences between movement-derived and base-generated dependencies stem from a difference between the featural triggers for external and internal Merge.

Keywords. A-resumption; broad subjects; multiple subjects; A-/Ā-distinction; feature-driven syntax; layering; parasitic gaps; secondary crossover

1 Introduction

The A-/Ā-distinction—a distinction between dependency types like raising and *wh*-movement—has been a mainstay of modern syntactic theorizing since at least Chomsky (1981). A variety of diagnostics have been shown to support this distinction (for recent overviews, see Richards 2014: sec. 1 and Safir 2019: sec. 2). For instance, raising does not induce (primary) weak crossover violations and cannot license parasitic gaps ((1)), while *wh*-movement displays the opposite profile ((2)).

(1) a. No puppy$_i$ seems [to its$_i$ friends] [___$_i$ to need a bath].
    b. * No puppy$_i$ seems [to everyone who plays with *pg*$_i$] [___$_i$ to need a bath].

(2) a. ?* Which puppy do [its$_i$ friends] think [___$_i$ needs a bath]?
    b. Which puppy does [everyone who plays with *pg*$_i$] think [___$_i$ needs a bath]?

Despite widespread acceptance of the empirical grounding of the A-/Ā-distinction, its theoretical underpinning remains a matter of debate. Traditional analyses made reference to the type of position heading the dependency: raising-to-subject lands in [Spec, TP], an A-position, while *wh*-movement lands in [Spec, CP], an Ā-position (see e.g. Chomsky 1981: 45, Mahajan 1990, Rizzi 1990, Chomsky 1993: 28–29, Chomsky and Lasnik 1993: 532, Tada 1993, Canac Marquis 1994, Merchant 1996, Keine 2018, Hewett 2023: ch. 7, 2025a, and Keine and Bhatt 2023). However, the position-based approach has been criticized for relying on stipulative (i.e. irreducible) classes of A- and Ā-positions.

As an alternative, several authors have attempted to derive the A-/Ā-distinction from properties of movement. Two types of movement-based approach can be identified in the prior literature.[1] The first type of

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[1]A third type of movement-based approach characterizes the A-/Ā-distinction as one between movements spanning different structural domains/distances: Ā-movement does, and A-movement does not, cross a transfer/Spell-Out domain (see Miyagawa 2010: esp. pp. 115–223 and Charnavel and Sportiche 2016: sec. 5.4.2). For reasons of space, I do not consider these approaches further.

approach posits a difference in the features responsible for A- versus Ā-movement. A prominent representative is van Urk (2015), who proposes that A-movement is driven by φ-features, whereas Ā-movement is driven by operator and discourse-related features like [wh] or [topic] (for conceptual predecessors, see Haegeman 1996 and Chomsky 2000: 108ff., 2007: 24, 2008: 150). van Urk's ideas have been adopted in much subsequent literature (e.g. Longenbaugh 2017, Bossi and Diercks 2019, Fong 2019, 2022, Wurmbrand 2019, Colley and Privoznov 2020, Coon et al. 2021, Scott 2021, Gong 2022, Lohninger et al. 2022, Chen 2023, Drummond 2023, Greeson 2023, Lohninger and Yip 2023, Ótott-Kovács 2023, Branan and Erlewine 2024, and Lohninger 2024).[2] The second movement-based approach is exemplified by Safir (2019), who argues that the A-/Ā-distinction is epiphenomenal and crucially related to the identity—and more specifically, the *category*—of the moving constituent. On Safir's analysis, constituents heading Ā-movement chains are categorially distinct from constituents heading A-movement chains.

Despite the success of movement-based approaches in accounting for contrasts like those in (1)–(2), I argue on the basis of novel data from Iraqi Arabic that any analysis which derives the A-/Ā-distinction from the mechanics of movement empirically undergenerates. This is because, as I will show, many of the familiar properties of A- and Ā-dependencies are also manifested in base-generated resumptive dependencies, which do not involve movement. Thus, one of the main takeaways of this paper is that, by investigating the properties of non-movement dependencies, we deepen our understanding of the underpinnings of the A-/Ā-distinction.

To that end, consider the distribution of resumptive pronouns in Iraqi Arabic (henceforth Iraqi). It is well known that Ā-dependencies can terminate in resumptive pronouns (in addition to gaps) in many languages, including *wh*-questions in Iraqi ((3)).

(3) jaː zoːlijja$_k$ tʃaːn Matt da-jxarrub {___$_k$ / -**ha**$_k$} min wusˤalti?
which carpet.FSG$_k$ AUX.PST.3MSG Matt PROG-destroy.3MSG {___$_k$ / -**3FSG**$_k$} from arrive.2FSG
(lit.) 'Which carpet$_k$ was Matt destroying {___$_k$ / it$_k$} when you arrived?' (Iraqi)

What is much less widely recognized, however, is the fact that A-dependencies can also terminate in resumptives—I refer to this phenomenon as *A-resumption* (to be contrasted with the more familiar *Ā-resumption*). For instance, in Iraqi, passive subjects of transitive verbs bind gaps ((4)), while subjects of pseudopassive sentences obligatorily bind resumptive pronouns ((5)). That the DP 'the carpet' heads an A-dependency in both examples is evinced by its triggering (third feminine singular) agreement on the past tense auxiliary located in T—a possibility distinctly unavailable to *wh*-phrases heading Ā-dependencies (cf. (6)) (I discuss agreement on both T and the lexical verb in greater detail in sections §3.1.1 and §5.1).

(4) [$_{TP}$ l-zoːlijja$_k$ tʃaːnat da-titxarrab ___$_k$].
[$_{TP}$ the-carpet.FSG$_k$ AUX.PST.3FSG PROG-destroy.PASS.3FSG ___$_k$]
'The carpet$_k$ was being destroyed ___$_k$.' (Iraqi)

(5) [$_{TP}$ l-zoːlijja$_k$ tʃaːnat *pro*$_{EXPL}$ da-jindaːs ʕaleː-**ha**$_k$].
[$_{TP}$ the-carpet.FSG$_k$ AUX.PST.3FSG *pro*$_{EXPL}$ PROG-step.PASS.3MSG on-**3FSG**$_k$]
(lit.) 'The carpet$_k$ was being stepped on it$_k$.' (Iraqi)

[2] For many, this includes Agree-related features under the assumption that Agree is a precondition on movement (on which see Chomsky 2000, 2001a). I do not make this assumption, though as we will see in sections §3.1.1 and §5.1, I maintain the idea that A- and Ā-dependencies differ with respect to how they interact with agreement.

Note additionally that some other authors derive the A-/Ā-distinction from differences between features that are not necessarily responsible for movement: Déprez (1989, 1994: esp. 132, (56)) presents an analysis that makes reference to Case (in addition to position types), while Borer (1995) derives the A-/Ā-distinction from properties of *chains*, which are defined according to the nature of the launching site of movement, the landing site of movement, and the moving element itself.

(6) * [$_{CP}$ jaː zoːlijja$_k$ ʧaːnat Matt da-jxarrub ___$_k$ min wusˤalti]?
[$_{CP}$ which carpet.FSG$_k$ AUX.PST.3FSG Matt PROG-destroy.3MSG ___$_k$ from arrive.2FSG]
(int.) 'Which carpet$_k$ was Matt destroying ___$_k$ when you arrived?' (Iraqi)

Various lines of evidence point to the conclusion that A-resumption in Iraqi is derived by base-generating a DP (e.g. 'the carpet' in (5)) in a non-thematic A-position, where it binds a resumptive pronoun.

Crucially, examining A-resumption sheds new light on the nature of the A-/Ā-distinction. I show—for the first time for several tests—that A-resumption simultaneously displays a mix of A- and Ā-properties. The table in (7) compares the properties of A-resumption with those of A-movement and of Ā-movement.

(7) *Comparison of the properties of A-movement, A-resumption, and Ā-movement in Iraqi*

	A-movement	**A-resumption**	**Ā-movement**
Can the head be agreed with by T?	✔	✔	✗
Can the head undergo A-movement?	✔	✔	✗
Can the head be controlled PRO?	✔	✔	–
Can the dependency license parasitic gaps?	✗	✗	✔
Does the dependency induce secondary WCO?	✗	✗	✔
Can the dependency bypass intervening DPs?	✗	✔	✔
Can the dependency span a finite clause?	✗	✔	✔

A-resumption and its mixed A-/Ā-behavior do not fit into traditional taxonomies which predict a strict bipartition between position and/or dependency types, viz. A vs. Ā. Additionally, as already suggested above, I will show that movement-based approaches to the A-/Ā-distinction, which largely derive mixed A-/Ā-properties from composite (or 'conjunctive') probing, cannot account for A-resumption for the simple reason that A-resumption does not involve movement. Nonetheless, I develop an analysis which preserves the insight from earlier work that *features* are at the heart of the A-/Ā-distinction.

In a nutshell, I propose that A-resumption is predicted by two hypothesized distinctions among feature types. The first distinction I draw is between A- and Ā-features, following the spirit of van Urk (2015). Specifically, building on ideas in Rezac (2003), Cable (2007, 2010), and Safir (2019) (and see recently Bošković 2024), I propose that the constituents heading A-dependencies are smaller than and categorially distinct from the constituents heading Ā-dependencies: the former are DPs, while the latter are QPs (where Q selects a complement, e.g. DP, to form QP). Thus, the A-/Ā-distinction can be reformulated in terms of categorial features as in (8).

(8) **A categorial view of the A-/Ā-distinction**
A-features target (features of) DPs. Ā-features target (features of) QPs.

The second distinction I draw, following Hewett (2023), is between the featural triggers for external and internal Merge. External Merge is driven by '•' (read: 'bullet') features (notation from Heck and Müller 2007, *et seq.*) and is responsible for building base-generated dependencies (e.g. resumption in Arabic). Internal Merge, by contrast, is driven by movement-specific '◁' (read: 'arrow') features.

By combining these two hypotheses about features (i.e. (i) D vs. Q and (ii) • vs. ◁), we arrive at the quadripartite distinction among feature types in (9).

(9) *Typology of features predicted by two hypothesized distinctions among features*

	A-feature	**Ā-feature**
◁-feature	[◁D◁] ↪ derives A-movement	[◁Q◁] ↪ derives Ā-movement
•-feature	[•D•] ↪ derives A-resumption	[•Q•] ↪ derives Ā-resumption

Every feature type is responsible for deriving a different dependency type, each with its own unique empirical profile. As I show, all are attested in Iraqi, supporting my feature-based approach. Moreover, I derive the apparently 'mixed' properties of A-resumption from independent properties of the structure-building feature [•D•].

The remainder of the paper is organized as follows. Section §2 lays out the basic properties of A-resumption in Arabic, focusing on Iraqi. Section §3 walks through the mixed A- and Ā-properties of A-resumption and compares A-resumption with A-movement and Ā-movement. In section §4, I develop an analysis of A- and Ā-dependencies based on two hypothesized distinctions among feature types: the first is a distinction between A- and Ā-features (D- and Q-features, respectively) and the second is a distinction between the features that trigger external and internal Merge. Section §5 then shows how the proposed feature system accounts for the A- and Ā-properties of A-resumption and other dependencies. Finally, section §6 concludes by summarizing the main theoretical and empirical contributions of the paper.

2 A-resumption in (Iraqi) Arabic: the basics of broad subjects

In Arabic, a DP can be merged in a non-thematic A-position, binding a resumptive pronoun.[3] In this paper, I focus on A-resumption in the verbal domain, though it is also attested in the adjectival domain in a construction the Arabic grammatical tradition calls the *naʿt sababī* 'indirect/semantically linked attribute' (see Cantarino 1975: 160–161; Polotsky 1978; Carter 1981: 245–247; Edzard 2013: sec. 7.4.4; and Badawi et al. 2015: 134–136, and references therein; for additional discussion of resumptive APs, see Hazout 2001, Kremers 2003: 99–102, Doron and Reintges 2006, Yoda 2014, and Youssef 2014). Based on evidence to be discussed in section §3, I contend that Arabic A-resumption is derived via a non-movement derivation involving base-generation of the A-binder in its high, non-thematic position, where it binds a resumptive pronoun (see also Doron 1996: sec. 4 and Doron and Heycock 1999: sec. 3).[4] This parallels the behavior of Ā-resumption in Arabic, which is derived by base-generating an Ā-binder in [Spec, CP] (Hewett 2023).

Much previous work has argued that varieties of Arabic permit clauses in which more than one DP bears subject-like properties; these include Jordanian (Hewett 2024a: 11–12, 2024b), Lebanese (Aoun et al. 2010: 229–235), Levantine (Alexopoulou et al. 2004), Modern Standard (Ayoub 1996: 747; Doron 1996; Doron and Heycock 1999, 2010; Heycock and Doron 2003; Alexopoulou et al. 2004; Aoun et al. 2010: 64–65, 229–235; Al-Aqarbeh and Al-Sarayreh 2017: 78; Tayalati and Danckaert 2020), Palestinian (Sellami 2024: esp. ch. 4), and Tunisian (Jlassi 2013: 161ff.; Hewett 2024b; Sellami 2024: esp. ch. 4). Following Doron

[3]A-resumption has also been argued to exist in other languages, including Akan (Yip and Ahenkorah 2023, 2024), Breton (Rezac 2011, 2013), Cantonese (Yip 2023; Yip and Ahenkorah 2023; but see Lai et al. 2024 for an alternative analysis of the Cantonese facts), Hebrew (Doron and Heycock 1999, 2010; Alexopoulou et al. 2004), Irish (McCloskey and Sells 1988; Maki and Ó Baoill 2008, 2011a,b; Ó Baoill 2013), and Samoan (Ershova 2023: 22). These authors differ, however, in whether or not they posit A-movement linking the head of the dependency with the resumptive pronoun. As I argue below, A-resumption in Iraqi is never generated by movement.

[4]Additional support for a non-movement derivation of A-resumption in Arabic comes from the fact that it is insensitive to coordinate structure islands, see Sellami (2024: 115, (4.40)) on Tunisian Arabic.

and Heycock (1999) *et seq.*, I refer to the lower, thematic subject as the *narrow subject* and to the higher, non-thematic subject as the *broad subject* (see Heycock 1993 and Yoon 2004, 2015 and references therein on the closely related topic of 'minor' and 'major' subjects in Japanese and Korean).[5] Broad subjects are semantically integrated into the clause by (obligatorily) binding a resumptive pronoun within their sister, the sentential predicate, via Predicate Abstraction (e.g. Heim and Kratzer 1998: 186, (4)).

Iraqi also permits broad subject constructions, as illustrated in (10): the broad subject 'the girls' binds the resumptive pronoun 'them' (*qua* variable) within the sentential predicate whose narrow subject is 'many rumors' via lambda abstraction.

(10) a. [$_{\text{TP}}$ [$_{\text{DP}}$ l-bnaːt]$_k$ ʧaːnaw [$_{\text{DP}}$ hwaːja iʃaːʕaːt] da-tinkitab ʕan-**hum**$_k$].
[$_{\text{TP}}$ $\underbrace{\text{[}_{\text{DP}}\text{ the-girls]}_k}_{\textit{broad subject}}$ AUX.PST.3PL $\underbrace{\text{[}_{\text{DP}}\text{ many rumors.FPL]}}_{\textit{narrow subject}}$ PROG-write.PASS.3FSG about-**them**$_k$]
'[The girls]$_k$ had many rumors being written about them$_k$.' (Iraqi)

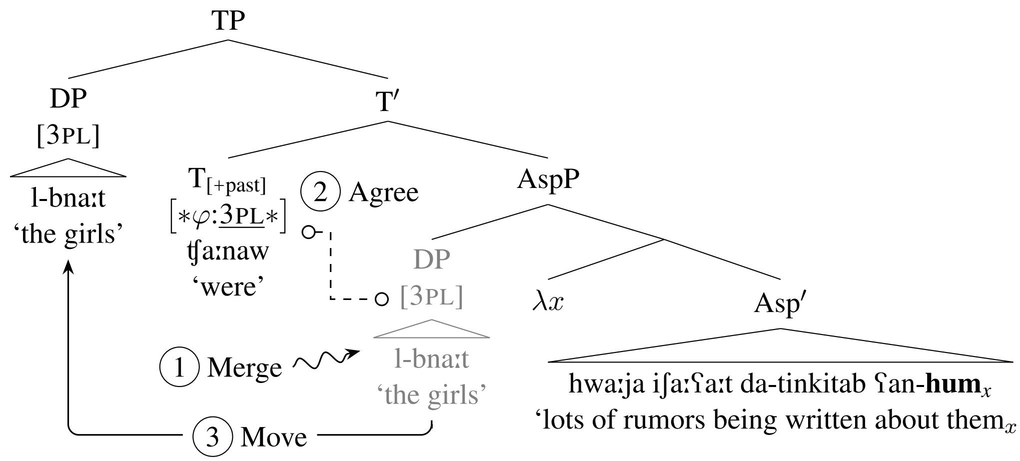
b. *Pseudo-LF of (10a)*
[$_{\text{TP}}$ [$_{\text{DP}}$ l-bnaːt] λx [… da-tinkitab ʕan-**hum**$_x$]].
[$_{\text{TP}}$ [$_{\text{DP}}$ the-girls] λx [… PROG-write.PASS.3FSG about-**them**$_x$]]

As the bracketing in (10) indicates, I contend with the work cited above that broad subjects are base-generated below the maximal projection of T as the heads of A-dependencies. While broad subjects might superficially resemble (clitic) left dislocated XPs, which occupy [Spec, CP] and head Ā-dependencies, several tests indicate that the two must be distinguished, *pace* Landau (2009, 2011), Alotaibi (2019), and Alqarni and Alanazi (2023). I will briefly lay out one decisive piece of evidence here and leave the remaining tests to be discussed in section §3.

Sellami (2024) observes that broad subjects can agree with an auxiliary (or copula) when the latter is present. Consider (10) once more: the broad subject *l-bnaːt* 'the girls' agrees in φ-features with the past tense auxiliary (which is a type of BE-auxiliary). The narrow subject *hwaːja iʃaːʕaːt* 'many rumors,' by contrast, triggers third feminine singular agreement on the main verb.[6] Assuming that BE-auxiliaries occupy T in Arabic (e.g. Benmamoun 2000; Soltan 2007, 2011; Aoun et al. 2010; Tucker 2011; Hallman 2015; Jarrah and Abusalim 2021), the broad subject must be base-generated close enough to T to be accessible for φ-probing. If Agree is only downward (see Deal To appear: 18 for references), then the broad subject must be generated below (i.e. in the c-command domain of) T. I propose that this position is a(n outer) specifier of Asp and that broad subjects may (but need not) subsequently move to [Spec, TP] (but see Sellami 2024 for a slightly different analysis which relies on a constrained form of Spec-Head agreement). The partial derivation in (11) illustrates how this works with the example in (10). I represent Agree-triggering features with '$*$' (read: 'star') diacritics, slightly adapted from the notation of Heck and Müller (2007) and Georgi (2017); unvalued features are written [$*$F:___$*$] and valued features are written [$*$F:$\underline{\text{VALUE}}$$*$].

[5] See Sellami (2024: esp. ch. 4) for evidence that Palestinian and Tunisian Arabic also permit DPs to be merged in low, non-thematic *object* positions; they dub such DPs *broad objects*.

[6] For the reader uninitiated in Arabic syntax, main verb agreement in (10a) manifests a pattern known as 'deflected' agreement where non-human plural DPs are cross-referenced by third person feminine singular morphology on verbs, adjectives, and pronouns, rather than the expected plural morphology (which is often unmarked for gender), regardless of the gender of the nominal in its singular form. For discussion, see Ferguson (1989), Belnap (1991), Hachimi (2011), Kramer and Winchester (2018), Bettega and D'Anna (2023), Hewett (2023: sec. 5.5.1), and Sellami (2024: sec. 5.4.3), as well as references cited in those works.

(11)


Crucially, XPs heading Ā-dependencies occupy [Spec, CP] and therefore are, by hypothesis, outside the search domain of a φ-probe on T. This explains why the heads of *wh*-dependencies fail to control auxiliary agreement: T must agree with the subject 'Matt' ((12a)) and cannot agree with the *wh*-phrase 'which carpet' in [Spec, CP] ((12b)).

(12) a. ✓ Agree

[$_{\text{CP}}$ jaː zoːlijja$_k$ ʧaːn Matt da-jxarrub ___$_k$ min wusˤalti]?
[$_{\text{CP}}$ which carpet.FSG$_k$ AUX.PST.3MSG Matt PROG-destroy.3MSG ___$_k$ from arrive.2FSG]
'Which carpet$_k$ was Matt destroying ___$_k$ when you arrived?' (Iraqi)

b. ✗ Agree

* [$_{\text{CP}}$ jaː zoːlijja$_k$ ʧaːnat Matt da-jxarrub ___$_k$ min wusˤalti]?
[$_{\text{CP}}$ which carpet.FSG$_k$ AUX.PST.3FSG Matt PROG-destroy.3MSG ___$_k$ from arrive.2FSG]
(int.) 'Which carpet$_k$ was Matt destroying ___$_k$ when you arrived?' (Iraqi)

I thus conclude that broad subjects head A-dependencies and are distinct from XPs heading Ā-dependencies. Throughout the paper, I rely on auxiliary agreement to diagnose the presence of broad subjects.

Before moving on, it is worth noting that Arabic is a subject pro-drop language (Eid 2011). As such, not all subjects are overt on the surface. This is relevant because many of the examples to be discussed employ expletive (or 'impersonal') pseudopassive sentences in which the narrow subject is a null expletive *pro* (i.e. *pro*$_{\text{EXPL}}$) bearing third masculine singular φ-features (see Mohammed 2000, Fassi Fehri 2009, and Tayalati and Danckaert 2020: sec. 5 on expletive *pro* in Arabic). Despite its covertness, *pro*$_{\text{EXPL}}$ can be detected via agreement. Example (13) illustrates with a typical Iraqi expletive pseudopassive sentence: *pro*$_{\text{EXPL}}$ is base-generated in a specifier of passive Voice (i.e. Voice[Pass])[7] and agrees with both the main verb, which moves at least as high as Asp (see e.g. Fassi Fehri 1993, Benmamoun 2000, Aoun et al. 2010, Tucker 2011, and Sellami 2024), and with an auxiliary in T if there is one.[8] For simplicity I do not show *pro*$_{\text{EXPL}}$ moving out of VoiceP, but nothing hinges on this.

(13) [$_{\text{TP}}$ ʧaːn [$_{\text{AspP}}$ da-jindaːs [$_{\text{VoiceP}}$ *pro*$_{\text{EXPL}}$ t_{V} ʕala l-zoːlijja]]].
[$_{\text{TP}}$ AUX.PST.3MSG [$_{\text{AspP}}$ PROG-step.PASS.3MSG [$_{\text{VoiceP}}$ $\underbrace{\textit{pro}_{\text{EXPL}}}_{\textit{narrow subject}}$ t_{V} on the-carpet.FSG]]]
'The carpet was being stepped on (lit. 'It was being stepped on the carpet.').' (Iraqi)

[7] This highly idiosyncratic distribution of *pro*$_{\text{EXPL}}$ may indicate that Voice[Pass] *lexically* selects (i.e. l-selects) *pro*$_{\text{EXPL}}$ (see Pesetsky 1991: 9–11; Merchant 2019; Zyman 2022: sec. 6, 2024: Supporting information; and Hewett 2024c).

[8] If VoiceP constitutes the clause-internal phase (at least in the passive, see Collins 2005: 98 and Hewett 2025c: 11–12), then my proposal that *pro*$_{\text{EXPL}}$ is externally merged in [Spec, VoiceP] in Iraqi has a close affinity to Deal (2009)'s analysis of the structural source of expletives in English (see also Bruening 2011).

Example (14) minimally differs from (13) by externally merging the broad subject 'the carpet' above pro_{EXPL}.[9] As expected, the broad subject agrees with the past tense auxiliary and binds a resumptive pronoun in its scope. Notably, pro_{EXPL} still controls agreement on the main verb, attesting to its presence in the clause.[10]

(14) [$_{\text{TP}}$ [$_{\text{DP}}$ l-zoːlijja]$_k$ ʧaːnat [$_{\text{AspP}}$ da-jindaːs pro_{EXPL} ʕaleː-**ha**$_k$]].
[$_{\text{TP}}$ [$_{\text{DP}}$ the-carpet.FSG]$_k$ AUX.PST.3FSG [$_{\text{AspP}}$ PROG-step.PASS.3MSG pro_{EXPL} on-**3FSG**$_k$]]
broad subject ... *narrow subject*
(lit.) 'The carpet$_k$ was being stepped on it$_k$.' (Iraqi)

Thus, I maintain that sentences like (14) ought to be analyzed as broad (or multiple) subject constructions, even though they only have a single *overt* subject.

3 A-resumption displays a mix of A- and Ā-properties

Given my claim that broad subjects occupy a non-thematic A-position in the clause, we might expect A-resumption to behave in all relevant respects like other A-dependencies such as A-movement. Yet, this expectation is not borne out. By contrasting A-resumption with A-movement and Ā-movement in Iraqi, I demonstrate that A-resumption simultaneously displays properties traditionally associated with both A-dependencies (section 3.1) and Ā-dependencies (section 3.2). The table in (15) summarizes my findings.

(15) *Comparison of the properties of A-movement, A-resumption, and Ā-movement in Iraqi*

	A-movement	**A-resumption**	**Ā-movement**
Can the head be agreed with by T?	✔	✔	✗
Can the head undergo A-movement?	✔	✔	✗
Can the head be controlled PRO?	✔	✔	–
Can the dependency license parasitic gaps?	✗	✗	✔
Does the dependency induce secondary WCO?	✗	✗	✔
Can the dependency bypass intervening DPs?	✗	✔	✔
Can the dependency span a finite clause?	✗	✔	✔

While I limit my discussion to Iraqi, parallel patterns of judgments hold for Jordanian Arabic (data omitted for space).

[9] Arabic does not permit P-stranding, ruling out true pseudopassives formed by A-movement of the complement of P.

[10] That sentences like (14) are truly passive can be shown by the fact that they permit an agentive 'by'-phrase headed by the preposition *min* 'from':

(i) [$_{\text{TP}}$ [$_{\text{DP}}$ l-zoːlijja]$_k$ ʧaːnat da-jindaːs pro_{EXPL} ʕaleː-**ha**$_k$ min Matt].
[$_{\text{TP}}$ [$_{\text{DP}}$ the-carpet.FSG]$_k$ AUX.PST.3FSG PROG-step.PASS.3MSG pro_{EXPL} on-**3FSG**$_k$ from Matt]
(lit.) 'The carpet$_k$ was being stepped on it$_k$ by Matt.' (Iraqi)

(ii) [$_{\text{TP}}$ [$_{\text{DP}}$ Joni]$_k$ ʧaːnat da-jinðˤiħak pro_{EXPL} ʕaleː-**ha**$_k$ min Matt].
[$_{\text{TP}}$ [$_{\text{DP}}$ Joni]$_k$ AUX.PST.3FSG PROG-laugh.PASS.3MSG pro_{EXPL} on-**3FSG**$_k$ from Matt]
'Joni$_k$ was being tricked by Matt (lit. 'Joni$_k$ was being laughed at her$_k$ by Matt').' (Iraqi)

Expletive pseudopassives in Iraqi thus differ from 'impersonal' sentences in various other languages which display *active* syntax, including prohibiting passive *by*-phrases (see e.g. Legate et al. 2020 and Šereikaitė 2021, 2022).

3.1 A-properties of A-resumption

I begin by laying out several A-properties of A-resumption.[11]

3.1.1 A-binders can agree with T

We have already encountered the first A-property of A-resumption—namely, its interaction with verbal agreement. I briefly review the facts here.

DPs that undergo A-movement in Iraqi—for instance, promoted passive subjects of transitive verbs—can agree with an auxiliary in T. In (16), the passive subject 'the carpet' agrees with the past tense auxiliary (and with the lexical verb).

(16) *A-movement can feed agreement with T*

A
[$_{\text{TP}}$ [$_{\text{DP}}$ l-zoːlijja]$_k$ ʧaːnat da-titxarrab ___$_k$].
[$_{\text{TP}}$ [$_{\text{DP}}$ the-carpet.FSG]$_k$ AUX.PST.3FSG PROG-destroy.PASS.3FSG ___$_k$]
'The carpet$_k$ was being destroyed ___$_k$.' (Iraqi)

By contrast, Ā-movement never feeds agreement with T in Iraqi. This is illustrated with *wh*-movement in (17): the *wh*-phrase 'which carpet' cannot agree with T—only the subject 'Matt' can.

(17) *Ā-movement cannot feed agreement with T*

Ā
[$_{\text{CP}}$ jaː zoːlijja$_k$ {ʧaːn / *ʧaːnat} Matt da-jxarrub ___$_k$]?
[$_{\text{CP}}$ which carpet.FSG$_k$ {AUX.PST.3MSG / *AUX.PST.3FSG} Matt PROG-destroy.3MSG ___$_k$]
'Which carpet$_k$ was Matt destroying ___$_k$?' (Iraqi)

As discussed in section 2, A-resumption patterns with A-movement in feeding agreement with T. In (18), the broad subject 'the carpet' agrees with the past tense auxiliary (see also Aoun et al. 2010: 230–231; Al-Aqarbeh and Al-Sarayreh 2017: 78; Wehbe 2023; and Sellami 2024).

(18) *A-resumption can feed agreement with T*

BIND
[$_{\text{TP}}$ [$_{\text{DP}}$ l-zoːlijja]$_k$ ʧaːnat da-jindaːs *pro*$_{\text{EXPL}}$ ʕaleː-**ha**$_k$].
[$_{\text{TP}}$ [$_{\text{DP}}$ the-carpet.FSG]$_k$ AUX.PST.3FSG PROG-step.PASS.3MSG *pro*$_{\text{EXPL}}$ on-**3FSG**$_k$]
broad subject *narrow subject*
(lit.) 'The carpet$_k$ was being stepped on it$_k$.' (Iraqi)

As Sellami (2024: 100–105) points out, this fact alone rules out an analysis in which broad subjects are base-generated in a left peripheral Ā-position, *pace* Landau (2009, 2011), Alotaibi (2019), and Alqarni and Alanazi (2023). Rather, broad subjects head A-dependencies and are accessible to φ-agreement by T.

[11] Doron (1996), Doron and Heycock (1999, 2010), Heycock and Doron (2003), and Alexopoulou et al. (2004) present a battery of additional diagnostics distinguishing broad subjects from (clitic) left dislocated XPs. These diagnostics, they contend, point to various other subject-like properties of broad subjects in Arabic (and Hebrew), including: (i) their ability to appear in a non-left-peripheral position, for instance to the right of auxiliaries, preposed sentential adjuncts, and fronted *wh*-phrases; (ii) their ability to appear in non-root contexts, such as within the antecedent of a conditional; (iii) their ability to be bare quantifiers (and, relatedly, to be information structurally neutral); and (iv) their ability to be ATB A-extracted out of a coordinate structure in which the parallel gap in the second conjunct is a narrow (rather than broad) subject position. For the sake of space, I do not discuss these properties of broad subjects further.

For various language-specific tests diagnosing the subjecthood of broad subjects in Breton, see Rezac (2011, 2013).

3.1.2 A-binders can undergo A-movement

The second A-property of A-resumption is its ability to feed A-movement. I focus on raising-to-subject (henceforth just ‘raising’), but see Doron (1996: 79–80), Doron and Heycock (1999: 71–72), and Alexopoulou et al. (2004: 335) on the ability of broad subjects to undergo raising-to-object in Standard and Levantine Arabic.[12]

DPs heading A-dependencies can undergo raising in Iraqi. In (19), the subject of the clausal complement of the raising verb *tˤilaʕ* ‘turn out, appear’—namely, *Joni*—raises to the matrix subject position.

(19) *Subjects can undergo raising-to-subject*

A

[$_{TP}$ tˤilʕat [$_{DP}$ Joni]$_k$ [___$_k$ da-tʃuːf Matt]].
[$_{TP}$ turn.out.3FSG [$_{DP}$ Joni]$_k$ [___$_k$ PROG-meet.3FSG Matt]]
‘Joni$_k$ turns out [___$_k$ to be meeting Matt].’ (Iraqi)

Note that such raising displays the hallmarks of A-movement, including feeding agreement (‘Joni’ controls agreement on the raising predicate ‘turn out’) and obeying A-minimality (to be discussed further in section §3.2.1)—that is, a DP cannot raise over an intervening DP. Consequently, only the embedded subject (*qua* highest DP) can raise into the matrix clause; an embedded object cannot raise over a subject:

(20) *Objects cannot undergo raising-to-subject over intervening subjects*

✗ A

*[$_{TP}$ tˤilʕat [$_{DP}$ Joni]$_k$ [[$_{DP}$ Matt] da-jʃuːf ___$_k$]].
[$_{TP}$ turn.out.3FSG [$_{DP}$ Joni]$_k$ [[$_{DP}$ Matt] PROG-meet.3MSG ___$_k$]]
(int.) ‘Joni$_k$ turns out [Matt to be meeting ___$_k$].’ (Iraqi)

Ā-movement, on the other hand, cannot feed A-movement—this is the classical Ban on Improper Movement (Chomsky 1973, 1981; May 1979; but cf. Kobayashi 2020). Wurmbrand (2019: 599, (26)) states (a more general version of) this ban in non-positional terms as a constraint on dependency types:

(21) *Improper A-after-Ā-constraint*
An A-dependency involving X cannot follow an Ā-dependency with X.

Thus, when attempting long-distance extraction of the *wh*-phrase ‘which girl’ in (22), once the *wh*-phrase has undergone Ā-movement to the edge of the embedded clause (or even to a low position in the higher clause), it cannot A-move to the subject position of the raising predicate ‘turn out’ and control matrix agreement.

(22) *Ā-moved* wh-*phrases cannot undergo raising-to-subject*

Ā ✗ A Ā

*[$_{CP}$ jaː bnajja$_k$ tˤilʕat ___$_k$ [___$_k$ [$_{DP}$ Matt] da-jʃuːf ___$_k$]].
[$_{CP}$ which girl$_k$ turn.out.3FSG ___$_k$ [___$_k$ [$_{DP}$ Matt] PROG-meet.3MSG ___$_k$]]
(int.) ‘Which girl$_k$ turns out [Matt to be meeting ___$_k$]?’ (Iraqi)

Of course, successive-cyclic Ā-movement is possible, though something else must fill the subject position of the raising predicate, such as a third masculine singular expletive *pro*$_{\text{EXPL}}$:[13]

[12] For evidence that A-binders in Irish A-resumption can undergo raising (either to subject or to the complement of a preposition), see McCloskey and Sells (1988: 157, (31) and 174–178) and Maki and Ó Baoill (2011b: sec. 2.1.2).

[13] I assume for explicitness that the expletive controls agreement on the matrix verb *tˤilaʕ* ‘turns out.’ However, it is also possible that the embedded clause itself is what controls main clause agreement. Resolving this issue is orthogonal to my analysis.

(23) Ā Ā
[$_{CP}$ jaː bnajja$_k$ *pro*$_{EXPL}$ tˤilaʕ [___$_k$ innu Matt da-jʃuːf ___$_k$]].
[$_{CP}$ which girl.F$_k$ *pro*$_{EXPL}$ turn.out.3MSG [___$_k$ that Matt PROG-meet.3MSG ___$_k$]]
'Which girl$_k$ does it turn out that [Matt is meeting ___$_k$]?' (Iraqi)

Strikingly, like A-movement and unlike Ā-movement, A-resumption can feed A-movement. Demonstrating this is complicated slightly by the fact that I have been unable to identify a raising predicate in Iraqi whose complement is large enough to contain an overt T auxiliary. While the lack of an agreeing auxiliary normally makes disambiguating between broad subjects and left dislocated XPs more challenging (which led to some previous work denying the existence of broad subjects, e.g. Landau 2009, 2011), the raising facts clearly distinguish the heads of A-dependencies from the heads of Ā-dependencies. Consider (24). The broad subject *Joni*, which binds a resumptive pronoun in the embedded clause, can undergo A-movement to the matrix subject position and control agreement on the raising predicate.

(24) *Broad subjects can undergo raising-to-subject*
A BIND
[$_{TP}$ tˤilʕat [$_{DP}$ Joni]$_k$ [___$_k$ da-jinðˤiħak *pro*$_{EXPL}$ ʕaleː-**ha**$_k$]].
[$_{TP}$ turn.out.3FSG [$_{DP}$ Joni]$_k$ [___$_k$ PROG-laugh.PASS.3MSG *pro*$_{EXPL}$ at-**3FSG**$_k$]]
(lit.) 'Joni$_k$ turns out [___$_k$ to be being made a fool of her$_k$].' (Iraqi)

Given that the Ban on Improper Movement is in force in Iraqi (see (22)), the acceptability of (24) suggests that broad subjects head A-dependencies, not Ā-dependencies.

This conclusion only follows, however, if the matrix subject moves from a position in the embedded clause. It is a priori possible that the subject of the raising verb could be base-generated in the matrix subject position, binding a resumptive pronoun in the embedded clause and circumventing A-movement altogether. I use scope reconstruction to rule out such a possibility. In (25), the raised broad subject 'a different girl' can take low scope with respect to the universally quantified temporal adverb 'every day', giving rise to a distributive reading.

(25) *Raised broad subjects can reconstruct for scope*
A BIND
[$_{TP}$ tˤilʕat [$_{DP}$ bnajja muxtilifa]$_k$ [___$_k$ da-jinðˤiħak *pro*$_{EXPL}$ ʕaleː-**ha**$_k$ kull yom min Matt]].
[$_{TP}$ turn.out.3FSG [$_{DP}$ girl.F different.F]$_k$ [___$_k$ PROG-laugh.PASS.3MSG *pro*$_{EXPL}$ on-**3FSG**$_k$ every day by Matt]]
(lit.) 'A different girl$_k$ turns out [___$_k$ to be being made a fool of her$_k$ every day by M.].' ($\forall > \exists$) (Iraqi)

To account for this inverse scope reading, I propose that 'a different girl' is reconstructed to the embedded broad subject position and that the quantificational adverb 'every day' undergoes clause-bound[14] QR to a position at the edge of the embedded clause c-commanding the broad subject.[15] I will note in passing that the availability of scope reconstruction distinguishes A-movement of broad subjects in Iraqi from copy raising constructions, which prohibit scope reconstruction (see Potsdam and Runner 2001: sec. 4.2 and Asudeh 2012: sec. 12.5 and references therein).

In summary, broad subjects can undergo raising, a property characteristic of the heads of A-dependencies.

[14]See Choueiri (2002: 101–108) on the absence of long-distance QR in (Lebanese) Arabic.

[15]Such clause-bound QR of quantificational adverbs, yielding inverse scope readings, is also available in monoclausal broad subject sentences.

3.1.3 A-binders can be controlled PRO

The third A-property—and more specifically, subject-like property—of A-binders in A-resumption is their ability to be controlled PRO (see Doron 1996: 81, Doron and Heycock 1999: 75–76 on Standard Arabic).[16]

Iraqi object control verbs like *nasˤsˤaħ* 'advise' in (26) select a controller DP and a clausal complement whose subject is controlled PRO.

(26) Misk$_i$ gaːlat [$_{CP}$ inn-i nasˤsˤaħt Joni$_k$ lbaːrħa [PRO$_{*i/k}$ tħasˤsˤil wðˤiːfa ʤdiːda]].
Misk.F$_i$ said.3FSG [$_{CP}$ that-1SG advised.1SG Joni.F$_k$ yesterday [PRO$_{*i/k}$ get.3FSG job new]]
'Misk$_i$ said that I advised Joni$_k$ yesterday [PRO$_{*i/k}$ to get a new job].'

Binding of PRO in Iraqi exhibits the familiar locality of obligatory control: the controller and the control clause must saturate or modify the same (control) predicate (see e.g. Landau 2024b: 2). Thus, *Joni* can bind the PRO subject of the control clause because *Joni* is an argument of *nasˤsˤaħ* and a co-argument of the control clause, but *Misk*, which is not an argument of *nasˤsˤaħ*, cannot bind PRO long-distance. Evidence that *Joni* is an argument of the control verb *nasˤsˤaħ* comes from the fact that it can be promoted to passive subject position like a typical object:

(27) Joni$_k$ innisˤħat ___$_k$ [PRO$_k$ tħasˤsˤil wðˤiːfa ʒdiːda].
Joni.F$_k$ advised.PASS.3FSG ___$_k$ [PRO$_k$ get.3FSG job new]
'Joni$_k$ was advised [PRO$_k$ to get a new job].'

As in other languages, PRO in Iraqi must be a subject. Thus, *Joni* cannot bind a PRO object in (28).

(28) * nasˤsˤaħt Joni$_k$ [Matt jʃuːf PRO$_k$].
advised.1SG Joni.F$_k$ [Matt.M hang.out.with.3MSG PRO$_k$]
(int.) 'I advised Joni$_k$ [Matt to hang out with PRO$_k$].'

I take the obligatory subjecthood of PRO for granted and do not attempt an explanation of this fact here (see Landau 2013: 108–115 for discussion). What is crucial is that broad subjects, like narrow subjects, can be controlled PRO. Consider (29).

(29) Misk$_i$ gaːlat [$_{CP}$ inn-i nasˤsˤaħt Joni$_k$ lbaːrħa [PRO$_{*i/k}$ jinqubaðˤ
Misk.F$_i$ said.3FSG [$_{CP}$ that-1SG advised.1SG Joni.F$_k$ yesterday [PRO$_{*i/k}$ arrest.PASS.3MSG
pro$_{\text{EXPL}}$ ʕaleː-**ha**$_{*i/k}$ min l-ʃurtˤa]].
pro$_{\text{EXPL}}$ on-**3FSG**$_{*i/k}$ by the-police]]
(lit.) 'Misk$_i$ said that I advised Joni$_k$ yesterday [PRO$_{*i/k}$ to get arrested (on) her$_{*i/k}$ by the police].'

This example manifests the same strict locality on obligatory control that we saw in (26): only *Joni*, which is a co-argument of the control clause, can control PRO. Nevertheless, Joni does not bind the narrow subject of the control clause, which is *pro*$_{\text{EXPL}}$, as indicated by the third masculine singular agreement on the passive verb. *Joni* is instead coindexed with the pronominal complement of the preposition *ʕala* 'on' (realized as *ʕaleː-* before pronominal clitics). Rather than abandon the cross-linguistically robust generalization that PRO must be a subject, we can account for control in (29) by positing a null PRO broad subject at the edge of the control clause. *Joni* thus binds the PRO broad subject and PRO A-binds the resumptive pronoun *-ha* 'her.' These facts are explainable only if broad subjects are sufficiently 'subject-like' to be realizable as PRO (e.g. they occupy the highest A-position in the embedded clause).

[16] McCloskey and Sells (1988) also argue that broad subjects in Irish can be controlled PRO, accounting for 'anomalous control' structures in which the controllee superficially appears to be the pronominal object of a preposition in the control clause (for additional data and discussion of the Irish facts, see Maki and Ó Baoill 2011b and Ó Baoill 2013).

3.1.4 A-dependencies do not license parasitic gaps

The fourth A-property of A-resumption is its inability to license parasitic gaps (*pg*s).

As originally observed by Engdahl (1983), A-movement does not license parasitic gaps (see van Urk 2017 for additional references and discussion). Thus, passive A-movement of the internal argument of a transitive verb to subject position in Iraqi does not license a parasitic gap in a clause-mate adjunct ((30)). For explicitness, I assume that such adjuncts adjoin to vP on the right, following Nissenbaum (2000).

(30) *A-movement does not license parasitic gaps*

A

[$_{\text{TP}}$ [$_{\text{DP}}$ haðoːli l-mumaθθilaːt]$_k$ twaðˁðˁafu [$_{\text{vP}}$ t_{V} ___$_k$] [biduːn ma aqaːbil {-hum$_k$/**pg*$_k$}]].
[$_{\text{TP}}$ [$_{\text{DP}}$ these the-actresses]$_k$ hired.PASS.3PL [$_{\text{vP}}$ t_{V} ___$_k$] [without C interview.1SG {-3PL$_k$/**pg*$_k$}]]
(lit.) 'These actresses$_k$ were hired ___$_k$ [without me interviewing {them$_k$/**pg*$_k$}].' (Iraqi)

Ā-movement of a *wh*-phrase, though, can license a parasitic gap, so long as other constraints on parasitic gap licensing are observed (e.g. the anti-c-command condition; see Hewett 2025c: sec. 3.3 for recent discussion and references). Example (31) shows that Ā-movement of an object *wh*-phrase to [Spec, CP] in Iraqi licenses a parasitic gap in a clause-mate vP-level adjunct (though the result is slightly degraded in line with the marked status of parasitic gaps in general).[17]

(31) *Ā-movement licenses parasitic gaps*

Ā

?[$_{\text{CP}}$ jaː mumaθθilaːt$_k$ waðˁðˁafti [$_{\text{vP}}$ t_{V} ___$_k$] [biduːn ma tqaːbiliːn *pg*$_k$]]?
[$_{\text{CP}}$ which actresses$_k$ hired.2FSG [$_{\text{vP}}$ t_{V} ___$_k$] [without C interview.2FSG *pg*$_k$]]
'Which actresses$_k$ did you hire ___$_k$ [without interviewing *pg*$_k$]?' (Iraqi; Hewett 2023: 62, (78b))

Relevant to the discussion at hand, A-resumption once more marches in lockstep with A-movement and does not license parasitic gaps. As the examples in (32) show, the binding dependency established between a broad subject and a resumptive pronoun fails to license a parasitic gap in a clause-mate vP-level adjunct.[18]

(32) *A-resumption does not license parasitic gaps*

a.
BIND
[$_{\text{TP}}$ [$_{\text{DP}}$ haðoːli l-mumaθθilaːt]$_k$ ʧaːnaw da-jinqubaðˁ [$_{\text{vP}}$ *pro*$_{\text{EXPL}}$ t_{V} ʕaleː-**hum**$_k$]
[$_{\text{TP}}$ [$_{\text{DP}}$ these the-actresses]$_k$ AUX.PST.3PL PROG-arrest.PASS.3MSG [$_{\text{vP}}$ *pro*$_{\text{EXPL}}$ t_{V} on-**3PL**$_k$]
[biduːn ma aqaːbil {-hum$_k$ / **pg*$_k$}]].
[without C interview.1SG {-3PL$_k$ / **pg*$_k$}]]
'These actresses$_k$ were being arrested [without me interviewing {them$_k$ / **pg*$_k$}].' (Iraqi)

b.
BIND
[$_{\text{TP}}$ [$_{\text{DP}}$ haðoːli l-banaːt]$_k$ ʧaːnaw [$_{\text{vP}}$ *pro*$_{\text{EXPL}}$ mbaːrak-**lhum**$_k$
[$_{\text{TP}}$ [$_{\text{DP}}$ these the-girls]$_k$ AUX.PST.3PL [$_{\text{vP}}$ *pro*$_{\text{EXPL}}$ congratulated.PASS.PART.MSG-**3PL.DAT**$_k$
b-naʤaːħ-hum$_k$] [gabl ma nʃuːf {-hum$_k$ / **pg*$_k$}]].
on-success-3PL$_k$] [before C see.1PL {-3PL$_k$ / **pg*$_k$}]]
'These girls$_k$ were congratulated on their success [before we saw {them$_k$ / **pg*$_k$}].' (Iraqi)

[17]For extensive discussion of parasitic gaps in Arabic *wh*-questions and relative clauses, see Hewett (2023: chs. 3.4–3.5), and see Alasmari (2024) on the distribution of parasitic gaps in Tehami Arabic focus fronting and clitic left dislocation.

[18]While I follow earlier work in taking finite verbs in Arabic to raise at least to Asp (see e.g. Fassi Fehri 1993; Benmamoun 2000; Soltan 2007, 2011; Aoun et al. 2010; Tucker 2011; Sellami 2024), I assume that participles such as *mbaːrak* 'congratulated.PASS.PART' in (32b), which are not inflected for person agreement or for tense, do not raise out of vP (see Hallman 2017 and Al-Raba'a 2021 on participles in Syrian and Jordanian Arabic, respectively). Nothing in my analysis hinges on this point.

In this respect, then, A-resumption straightforwardly behaves like an A-dependency.

3.1.5 A-binders can indirectly bind pronouns (i.e. they do not induce secondary weak crossover)

The final A-property of A-resumption that I will discuss is its failure to induce secondary weak crossover violations. Before delving into the A-resumption facts, I provide a short primer on crossover effects.

Crossover (Postal 1971) describes the failure of certain pronouns that I will call *crossed pronouns* to be bound by the heads of Ā-dependencies in specific structural configurations. Previous literature has identified at least two variables relevant for categorizing types of crossover.[19] The first is the strong vs. weak distinction introduced by Wasow (1972): crossover is strong when the Ā-bound variable (often, though not always, a gap left by Ā-movement, see Hewett 2023: ch. 7, 2025a for discussion) c-commands the crossed pronoun and it is weak when neither the Ā-bound variable nor the crossed pronoun c-commands the other. Strong and weak crossover effects are illustrated with Iraqi *wh*-questions in the following examples (I abstract away from intermediate landing sites of Ā-movement for simplicity):

(33) *Ā-movement induces (primary) strong crossover effects*

Ā

*[CP minu$_k$ titwaqqaʕiːn *pro*$_k$ jriːd Hend tixtaːr ___$_k$]?
[CP who$_k$ think.2FSG *pro*$_k$ wants.3MSG Hend choose.3FSG ___$_k$]
(int.) 'Who$_k$ do you think he$_k$ wants Hend to choose ___$_k$?' (Iraqi; Hewett 2023: 333, (9))

(34) *Ā-movement induces (primary) weak crossover effects*

Ā

*[CP minu$_k$ titwaqqaʕiːn [umm-a$_k$] triːd Hend tixtaːr ___$_k$]?
[CP who$_k$ think.2FSG [mother-his$_k$] wants.3FSG Hend choose.3FSG ___$_k$]
(int.) 'Who$_k$ do you think [his$_k$ mother] wants Hend to choose ___$_k$?' (Iraqi; Hewett 2023: 333, (8))

Both examples instantiate crossover because a pronoun (*pro* 'he' in (33), *-a* 'his' in (34)) fails to be bound by the Ā-moved *wh*-phrase *minu* 'who' from its landing site. Moreover, (33) involves strong crossover because the crossed pronoun *pro* c-commands the variable site while (34) involves weak crossover because the crossed pronoun *-a* 'his' does not c-command the variable site.

The second distinction relevant for analyzing crossover effects is that between primary and secondary crossover (e.g. Postal 1993, Safir 1996, Keine and Bhatt 2023, and Hewett 2025a; see Hewett 2023: 331 for additional references). Primary crossover effects arise when the crossed pronoun and the Ā-bound variable bear the same index, as in (33)–(34). Secondary crossover, on the other hand, arises when the crossed pronoun and the Ā-bound variable bear different indices due to pied piping. In a nutshell, secondary crossover describes the failure of a constituent properly contained within the head of an Ā-dependency to bind out of its container phrase. I henceforth refer to such binding out of a container phrase (i.e. binding in the apparent absence of c-command) as *indirect binding* (Büring 2004; Hewett 2023: ch. 7). The following examples illustrate secondary strong and weak crossover with Iraqi *wh*-questions:

(35) *Ā-movement induces secondary strong crossover effects*

[19] I set aside cases of so-called *weakest crossover* where expected crossover effects are missing (Lasnik and Stowell 1991; Postal 1993); see Ruys (2004) and Safir (2004: sec. 3.6) for two possible analyses of these facts.

Ā

✗ INDIRECT BIND

*[$_{\text{CP}}$ [sˤadiːqat minu$_i$]$_k$ taʕtaqidiːn *pro*$_i$ raħ jixtaːr ___$_k$ li-l-liʕba]?
[$_{\text{CP}}$ [friend.FSG who$_i$]$_k$ suspect.2FSG *pro*$_i$ FUT choose.3MSG ___$_k$ for-the-game]
(int.) '[Whose$_i$ friend]$_k$ do you suspect he$_i$ will choose ___$_k$ for the game?' (Iraqi; Hewett 2023: 360, (39ai))

(36) *Ā-movement induces secondary weak crossover effects*

Ā

✗ INDIRECT BIND

*[$_{\text{CP}}$ [ʔasˤdiqaːʔ minu$_i$]$_k$ taʕtaqidiːn sˤaːħibt-a$_i$ raħ tixtaːr ___$_k$ li-l-liʕba]?
[$_{\text{CP}}$ [friends who$_i$]$_k$ suspect.2FSG girlfriend-his$_i$ FUT choose.3FSG ___$_k$ for-the-game]
(int.) '[Whose$_i$ friends]$_k$ do you suspect his$_i$ girlfriend will choose ___$_k$ for the game?' (Iraqi; Hewett 2023: 361, (40ai))

Example (35) illustrates secondary crossover because the crossed pronoun and gap are contra-indexed—the pronoun is co-indexed with the embedded *wh*-phrase *minu* 'who(se)' and the gap with the entire pied-piped phrase *sˤadiːqat minu* 'whose friend'—and this crossover is strong because the crossed pronoun c-commands the (contra-indexed) Ā-bound gap. Example (36) similarly involves secondary crossover, though in this instance crossover is weak because the crossed pronoun does not c-command the Ā-bound gap.

Typically, the literature on the A-/Ā-distinction has focused on whether or not a given dependency triggers *primary* weak crossover effects. However, Hewett (2023: ch. 7, 2025a) identifies a problematic ambiguity in diagnosing primary crossover effects in resumptive dependencies (see also Korsah and Murphy 2024: sec. 3.4): it is often not possible to distinguish the crossed pronoun from the resumptive pronoun, especially in languages like Arabic where resumptive pronouns are formally indistinct from regular pronouns.[20] For that reason, Hewett devotes most of his attention to *secondary* crossover effects in his investigation of Ā-resumption, and I will follow suit here, focusing specifically on secondary weak crossover.

Returning to the A-/Ā-distinction, while Ā-dependencies regularly trigger secondary weak crossover (for movement, see (36); for base-generated resumption, see Hewett 2023: sec. 7.4), A-dependencies are immune from such effects (e.g. Büring 2005: 244–246). Unfortunately, the latter fact cannot be demonstrated in Iraqi, as I am unaware of any kind of A-movement in Iraqi that allows one DP to skip over another (or over an intervening PP) (see McGinnis 1998 and Keine and Zeijlstra 2023 for discussion of such leap-frogging movement). Nonetheless, it is easily demonstrated with English raising-to-subject. In (37), A-movement of *every girl's bag* over the experiencer PP *to her friends* does not trigger a secondary weak crossover effect: *every girl* can indirectly bind *her* after its container phrase has undergone A-movement to a position c-commanding *her*.

(37) *(English) A-movement does not induce secondary weak crossover effects*

A

INDIRECT BIND

[$_{\text{TP}}$ [$_{\text{DP}}$ Every girl$_i$'s bag]$_k$ seems to [her$_i$ friends] [$_{\text{TP}}$ ___$_k$ to have her name written on it]].

What's more, (38) shows that the high A-position of *every girl's bag* is crucial: without raising, indirect binding is impossible.[21]

(38) *[$_{\text{TP}}$ It seems to [her$_i$ friends] [$_{\text{CP}}$ that [$_{\text{DP}}$ every girl$_i$'s bag] has her name written on it]].

[20] Hewett (2023: 4–5) refers to the morphophonological isomorphism between ordinary pronouns and resumptive pronouns as the *Doron–Engdahl–McCloskey Generalization*; see that work for references and discussion.

[21] Attempted QR of the embedded quantifier phrase to a position c-commanding the bound pronoun would also induce crossover, see Chomsky (1976: esp. 340–343) and Büring (2005: 165–166).

Crucially, A-resumption behaves like A-movement for the purposes of evaluating crossover, as shown by the Iraqi example in (39): a quantified possessor (i.e. 'every girl') embedded within a broad subject (i.e. 'every girl's bag') can indirectly bind a pronoun (i.e. 'her') that neither c-commands nor is c-commanded by the resumptive pronoun in the variable site.

(39) *A-resumption does not induce secondary weak crossover effects*

BIND
INDIRECT BIND

[$_{\text{TP}}$ [$_{\text{DP}}$ ʤantˁat kull bnajja$_i$]$_k$ ʧaːnat [ism-ha$_i$] da-jinkitab ʕaleː-**ha**$_k$].
[$_{\text{TP}}$ [$_{\text{DP}}$ bag.FSG every girl$_i$]$_k$ AUX.PST.3FSG [name.MSG-her$_i$] PROG-write.PASS.3MSG on-**3FSG**$_k$]
'[Every girl$_i$'s bag]$_k$ had [her$_i$ name] being written on it$_k$.' (Iraqi)

As in English, indirect binding in Iraqi is impossible if the container phrase does not occupy an A-position that c-commands the crossed pronoun:

(40) * [$_{\text{TP}}$ [ism-ha$_i$] ʧaːn da-jinkitab ʕala [$_{\text{DP}}$ ʤantˁat kull bnajja$_i$]].
[$_{\text{TP}}$ [name.MSG-her$_i$] AUX.PST.3MSG PROG-write.PASS.3MSG on [$_{\text{DP}}$ bag.FSG every girl$_i$]]
(int.) '[Her$_i$ name] was being written on [every girl$_i$'s bag].' (Iraqi)

In summary, indirect binding of pronouns is possible from the heads of both A-movement and A-resumption dependencies, but not from the heads of Ā-dependencies. This discovery provides additional strong evidence in support of the view that broad subjects head A-dependencies.

The results of my investigation into the contrasting properties of A-movement, A-resumption, and Ā-movement up to this point are summarized in the following table:

(41) *Comparison of the properties of A-movement, A-resumption, and Ā-movement in Iraqi* (to be completed)

	A-movement	**A-resumption**	**Ā-movement**
Can the head of the dependency be agreed with by T?	✔	✔	✗
Can the head of the dependency undergo A-movement?	✔	✔	✗
Can the head of the dependency be controlled PRO?	✔	✔	–
Can the dependency license parasitic gaps?	✗	✗	✔
Does the dependency induce secondary WCO?	✗	✗	✔

3.2 Ā-properties of A-resumption

Despite the battery of arguments put forward in the previous section in favor of analyzing A-resumption as an A-dependency, the behavior of A-resumption diverges from that of A-movement in at least two respects: unlike A-movement and like Ā-movement,[22] A-resumption does not display DP-minimality effects (section §3.2.1) or clause-boundedness (section §3.2.2).

[22] And also like Ā-resumption, which does not display DP-minimality or clause-boundedness.

3.2.1 A-resumption can skip intervening DPs

A-movement cannot skip intervening DPs in Iraqi.[23] This was shown for raising-to-subject in section §3.1.2: only the highest DP in the clausal complement of a raising verb can undergo raising-to-subject (see especially example (20)). In this section, I demonstrate that passive A-movement in Iraqi also manifests DP-relativized minimality effects.

Iraqi can form double object constructions, which contain two DP internal arguments, with verbs like *minaħ* 'award' ((42)). In the passive, only the indirect object DP can undergo A-movement to the canonical subject position ((43a)); the direct object cannot A-move over the indirect object and control agreement[24] on the verb ((43b)). Thus, Iraqi only permits 'asymmetric', rather than 'symmetric', passives, and specifically only indirect object passives (see Citko et al. 2017: sec. 4.1 and Holmberg et al. 2019: sec. 1 and references therein on cross-linguistic variation in this domain).

(42) [$_{\text{TP}}$ minħit [$_{\text{DP}}$ l-tˤullaːb] [$_{\text{DP}}$ l-fiːzaːt]].
[$_{\text{TP}}$ awarded.1SG [$_{\text{DP}}$ the-students.MPL] [$_{\text{DP}}$ the-visas.FPL]]
'I awarded the students the visas.' (Iraqi)

(43) *A-movement cannot skip intervening DPs*

A

a. [$_{\text{TP}}$ [$_{\text{DP}}$ l-tˤullaːb]$_k$ inminħaw ___$_k$ [$_{\text{DP}}$ l-fiːzaːt]].
[$_{\text{TP}}$ [$_{\text{DP}}$ the-students.MPL]$_k$ awarded.PASS.3PL ___$_k$ [$_{\text{DP}}$ the-visas.FPL]]
'The students$_k$ were awarded ___$_k$ the visas.' (Iraqi)

✗ A

b. * [$_{\text{TP}}$ [$_{\text{DP}}$ l-fiːzaːt]$_k$ inminħat [$_{\text{DP}}$ l-tˤullaːb] ___$_k$].
[$_{\text{TP}}$ [$_{\text{DP}}$ the-visas.FPL]$_k$ awarded.PASS.3FSG [$_{\text{DP}}$ the-students.MPL] ___$_k$]
(int.) 'The visas$_k$ were awarded the students ___$_k$.' (Iraqi)

This contrasts sharply with the behavior of Ā-movement, which freely skips over non-*wh* DPs: in (44), *wh*-movement of the direct object *jaː zoːlijja* 'which carpet' crosses the DP subject *Matt* without issue.

(44) *Ā-movement can skip intervening (non-*wh*) DPs*

Ā

[$_{\text{CP}}$ jaː zoːlijja$_k$ ʧaːn [$_{\text{DP}}$ Matt] da-jxarrub ___$_k$ min wusˤalti]?
[$_{\text{CP}}$ which carpet.FSG$_k$ AUX.PST.3MSG [$_{\text{DP}}$ Matt] PROG-destroy.3MSG ___$_k$ from arrive.2FSG]
'Which carpet$_k$ was Matt destroying ___$_k$ when you arrived?' (Iraqi)

A-resumption patterns with Ā-movement in this respect. As (45a)–(45b) show, a broad subject ('these girls' and 'this girl', respectively) can A-bind a resumptive pronoun over an intervening narrow subject DP ('many rumors' and 'few people', respectively). DPs situated along the dependency path do not trigger violations of minimality under A-resumption.[25]

(45) *A-resumption can skip intervening DPs*

[23] This is not to imply that A-movement never skips intervening DPs cross-linguistically. In fact, there are good arguments in favor of 'leap-frogging' A-movement in many languages (see McGinnis 1998; Legate 2014; and Keine and Zeijlstra 2023). Nevertheless, Iraqi clearly lacks such leap-frogging A-movement, at least in raising-to-subject and passivization.

[24] Which we would expect to be *deflected*, i.e. third feminine singular; see footnote 6.

[25] For examples showing a lack of minimality in Syrian Arabic resumptive passive participles, see Cowell (1964: 264, fn. 2).

a. [TP [DP haðoːli l-bnaːt]$_k$ tʃaːnaw [DP hwaːja iʃaːʕaːt] da-tinkitab ʕan-**hum**$_k$
[TP [DP these the-girls]$_k$ AUX.PST.3PL [DP many rumors.FPL] PROG-write.PASS.3FSG about-**them**$_k$
b-l-dʒariːda].
in-the-newspaper]
'These girls$_k$ had many rumors name being written about them$_k$ in the newspaper.' (Iraqi)

b. [TP [DP haːj l-bnajja]$_k$ tʃaːnat [DP tʃam ʃaxisˤ] da-jsallimuːn ʕaleː-**ha**$_k$
[TP [DP this.FSG the-girl.F]$_k$ AUX.PST.3FSG [DP few person] PROG-shake.hands.3PL on-**3FSG**$_k$
min wusˤalna].
from arrive.1PL]
'This girl had a few people shaking hands with her when we arrived.' (Iraqi)

This is the first indication that A-resumption in Iraqi does not simply involve run-of-the-mill A-movement from the variable site (see also Doron 1996: sec. 4 and Doron and Heycock 1999: sec. 3), *pace* Wehbe (2023).[26] Contrast, for instance, analyses of copy raising constructions according to which a 'resumptive' pronoun accompanies regular A-movement (e.g. Fujii 2007).[27]

3.2.2 A-resumption can span finite clause boundaries

The second way in which A-resumption does not behave like a typical A-dependency is in its locality profile.

A-movement in Iraqi cannot cross finite clause boundaries.[28] As examples (46a)–(46b) illustrate, attempted A-movement of an object DP out of a finite CP and into the subject position of a higher clause (as diagnosed by matrix copular and verbal agreement) yields unacceptability in Iraqi.

[26]A-resumption also does not exhibit A-over-A minimality effects, contrary to what would be expected if A-resumption simply involved A-movement: a broad subject can bind a resumptive pronoun that is properly contained in a distinct DP ((i)) or in a coordination of two (or more) DPs ((ii)).

(i) Raːnia$_k$ keːnət [DP ʃʕar-**ha**$_k$] jətˤweːl fissaʕ.
Rania.F$_k$ AUX.PST.3FSG [DP hair.MSG-**3FSG**] be.long.3MSG quick
'Rania's hair used to grow quickly.' (Tunisian; slightly adapted from Sellami 2024: 115, fn. 14, (i))

(ii) Raːnia$_k$ keːnət [&P **hijja**$_k$ w-ʕAzza] jʒiːw l-əl-faːk koll nhaːr.
Rania.F$_k$ AUX.PST.3FSG [&P **3FSG**$_k$ and-Azza] come.3PL to-the-university every day
'Rania$_k$, she$_k$ and Azza used to come to campus every day.' (Tunisian; slightly adapted from Sellami 2024: 115, (4.40))

For additional examples, see Sellami (2024: ch. 4) on Tunisian and Palestinian Arabic, Aoun et al. (2010: 231, (41b)) on Standard Arabic, and Wehbe (2023) on Lebanese Arabic.

[27]Relatedly, see Hewett (2023) for arguments that Arabic resumptive Ā-dependencies never involve Ā-movement, contra Aoun (2000), Aoun et al. (2001), Choueiri (2002), Aoun and Li (2003), Demirdache and Percus (2009, 2011), and Sichel (2014).

[28]This section focuses on A- and Ā-movement of embedded objects. While some Arabic varieties have been shown to permit A-movement of embedded subjects out of finite CPs (i.e. hyperraising-to-subject, see Farghal 2020 and Camilleri and Sadler 2023), Iraqi seems to forbid such A-movement, at least with predicates like 'be {known, suspected}.' In (i), the matrix subject 'a different girl' cannot take scope below the universally quantified embedded object DP 'every car,' despite the fact that direct objects can scope over clausemate subjects in Arabic.

✗ A

(i) #[TP [DP bnajja muxtilifa]$_k$ tʃaːnat maʃkuːk [CP innu ___$_k$ baːgat kull sajjaːra]].
[TP [DP girl.F different.F]$_k$ was.3FSG suspected.MSG [CP that ___$_k$ stole.3FSG every car.F]]
(int.) 'A different girl was suspected that ___$_k$ stole every car.' ($\exists > \forall$, $*\forall > \exists$) (Iraqi)

The lack of an inverse scope reading in (i) is explainable if Iraqi does not allow hyperraising-to-subject. On the other hand, a surface scope reading is possible because 'a different girl' can be base-generated in the matrix clause as a broad subject, binding a *pro* resumptive subject in the embedded clause.

(46) *A-movement cannot cross finite clause boundaries*

✗ A

a. *[$_{\text{TP}}$ [$_{\text{DP}}$ haðoːli l-banaːt]$_k$ ʧaːnaw maʕruːfaːt [$_{\text{CP}}$ innu kull l-naːs jħibbuːn ___$_k$]].
[$_{\text{TP}}$ [$_{\text{DP}}$ these the-girls]$_k$ were.3PL known.FPL [$_{\text{CP}}$ that all the-people like.3PL ___$_k$]]
(int.) 'These girls$_k$ were known that everybody likes ___$_k$ (∼ to be liked by everybody).' (Iraqi)

✗ A

b. *[$_{\text{TP}}$ [$_{\text{DP}}$ haːj l-sajjaːra]$_k$ ʧaːnat maʃkuːka [$_{\text{CP}}$ innu bigti ___$_k$]].
[$_{\text{TP}}$ [$_{\text{DP}}$ this.FSG the-car.F]$_k$ was.3FSG suspected.FSG [$_{\text{CP}}$ that stole.2FSG ___$_k$]]
(int.) 'This car$_k$ was suspected that you stole ___$_k$ (∼ to have been stolen by you).' (Iraqi)

Note that the agreement on the matrix lexical verb (here, a passive participle) is orthogonal to the ungrammaticality of such cross-clausal A-movement. Even if the lexical verb bears masculine singular features, which I analyze as the result of agreement with an expletive *pro*$_{\text{EXPL}}$,[29] the examples remain unacceptable:

(47) a. *[$_{\text{TP}}$ [$_{\text{DP}}$ haðoːli l-banaːt]$_k$ ʧaːnaw *pro*$_{\text{EXPL}}$ maʕruːf [$_{\text{CP}}$ innu kull l-naːs jħibbuːn ___$_k$]].
[$_{\text{TP}}$ [$_{\text{DP}}$ these the-girls]$_k$ were.3PL *pro*$_{\text{EXPL}}$ known.MSG [$_{\text{CP}}$ that all the-people like.3PL ___$_k$]]
(int.) 'These girls were known to be liked by everybody.' (Iraqi)

b. *[$_{\text{TP}}$ [$_{\text{DP}}$ haːj l-sajjaːra]$_k$ ʧaːnat *pro*$_{\text{EXPL}}$ maʃkuːk [$_{\text{CP}}$ innu bigti ___$_k$]].
[$_{\text{TP}}$ [$_{\text{DP}}$ this.FSG the-car.F]$_k$ was.3FSG *pro*$_{\text{EXPL}}$ suspected.MSG [$_{\text{CP}}$ that stole.2FSG ___$_k$]]
(int.) 'This car was suspected to have been stolen by you.' (Iraqi)

By contrast, Ā-movement in *wh*-questions freely crosses finite CP boundaries:

(48) *Ā-movement can cross finite clause boundaries*

Ā

a. [$_{\text{CP}}$ jaː bnaːt$_k$ ʧaːn *pro*$_{\text{EXPL}}$ maʕruːf [$_{\text{CP}}$ innu kull l-naːs jħibbuːn ___$_k$]]?
[$_{\text{CP}}$ which girls$_k$ was.3MSG *pro*$_{\text{EXPL}}$ known.MSG [$_{\text{CP}}$ that all the-people like.3PL ___$_k$]]
'Which girls$_k$ was it known that everybody likes ___$_k$?' (Iraqi)

Ā

b. [$_{\text{CP}}$ jaː sajjaːra$_k$ ʧaːn *pro*$_{\text{EXPL}}$ maʃkuːk [$_{\text{CP}}$ innu bigti ___$_k$]].
[$_{\text{TP}}$ which car.F$_k$ was.3MSG *pro*$_{\text{EXPL}}$ suspected.MSG [$_{\text{CP}}$ that stole.2FSG ___$_k$]]
'Which car$_k$ was it suspected that you stole ___$_k$?' (Iraqi)

As the examples in (49) illustrate, A-resumption displays the more permissive locality profile of an Ā-dependency: a broad subject can bind a resumptive pronoun across a finite CP boundary.[30,31]

[29] Alternatively, the third masculine singular agreement could result from agreement with the embedded CP.

[30] The matrix lexical verb, however, cannot agree with the broad subject:

(i) * [$_{\text{TP}}$ [$_{\text{DP}}$ haðoːli l-banaːt]$_k$ ʧaːnaw **maʕruːfaːt** [$_{\text{CP}}$ innu kull l-naːs jħibbuː-hum$_k$]].
[$_{\text{TP}}$ [$_{\text{DP}}$ these the-girls.F]$_k$ were.3PL **known.FPL** [$_{\text{CP}}$ that all the-people like.3PL-3PL$_k$]]
(int.) 'These girls$_k$ were known that everybody likes them$_k$ (∼ to be liked by everybody).' (Iraqi)

(ii) * [$_{\text{TP}}$ [$_{\text{DP}}$ haːj l-sajjaːra]$_k$ ʧaːnat **maʃkuːka** [$_{\text{CP}}$ innu bigtiː-ha$_k$]].
[$_{\text{TP}}$ [$_{\text{DP}}$ this.FSG the-car.F]$_k$ was.3FSG **suspected.FSG** [$_{\text{CP}}$ that stole.2FSG-3FSG$_k$]]
(int.) 'This car$_k$ was suspected that you stole it$_k$ (∼ to have been stolen by you).' (Iraqi)

This follows from the expected interaction between the directionality of Agree, the distribution of φ-probes in the clause, and the base-generation site of broad subjects. Specifically, if Agree is only downward (e.g. Deal To appear: 18 and references therein), if Arabic main verb agreement is located on Asp or a similar head below T in the extended verbal projection (e.g. Soltan 2007, 2011; Sellami 2024), and if broad subjects are base-generated in [Spec, AspP] (see section §2), then a broad subject will never be in the search domain of a main verb agreement φ-probe.

[31] See Hazout (2001: 104–105) and Belikova (2008: 14) for a similar finding for A-resumption in Standard Arabic participles.

(49) *A-resumption can cross finite clause boundaries*

a. [TP [DP haðoːli l-banaːt]$_k$ ʧaːnaw pro_{EXPL} maʕruːf [CP innu kull l-naːs jħibbuː-**hum**$_k$]].
[TP [DP these the-girls]$_k$ were.3PL pro_{EXPL} known.MSG [CP that all the-people like.3PL-**3PL**$_k$]]
(lit.) 'These girls$_k$ were known that everybody likes them$_k$ (∼ to be liked by everybody).' (Iraqi)

b. [TP [DP haːj l-sajjaːra]$_k$ ʧaːnat pro_{EXPL} maʃkuːk [CP innu bigtiː-**ha**$_k$]].
[TP [DP this.FSG the-car.F]$_k$ was.3FSG pro_{EXPL} suspected.MSG [CP that stole.2FSG-**3FSG**$_k$]]
(lit.) 'This car$_k$ was suspected that you stole it$_k$ (∼ to have been stolen by you).' (Iraqi)

Such examples are typically judged to be somewhat marked, but they are nevertheless acceptable, and, crucially, much improved over their A-movement–derived gapped counterparts (e.g. (47a)–(47b)).

The complete results of our investigation into the properties of A-movement, A-resumption, and Ā-movement are presented in (50).

(50) *Comparison of the properties of A-movement, A-resumption, and Ā-movement in Iraqi* (completed)

	A-movement	**A-resumption**	**Ā-movement**
Can the head be agreed with by T?	✔	✔	✗
Can the head undergo A-movement?	✔	✔	✗
Can the head be controlled PRO?	✔	✔	–
Can the dependency license parasitic gaps?	✗	✗	✔
Does the dependency induce secondary WCO?	✗	✗	✔
Can the dependency bypass intervening DPs?	✗	✔	✔
Can the dependency span a finite clause?	✗	✔	✔

While these properties consistently distinguish A-movement from Ā-movement in Iraqi, A-resumption presents something of a mixed bag. It is the goal of the next two sections to explain why that is.

4 A feature-driven analysis of A- and Ā-dependencies

In this section, I develop an analysis of the features that build A- and Ā-dependencies and I leverage this analysis in section §5 to explain the mixed properties of A-resumption. My analysis of the A-/Ā-distinction is summarized in the conjecture in (51), which is closely modeled after a proposal in van Urk (2015: 26).[32]

(51) **A featural approach to the A-/Ā-distinction**
All differences between A- and Ā- dependencies derive from the features which construct those dependencies.

[32]There are at least two important differences between van Urk's featural approach to the A-/Ā-distinction and my own. First, van Urk's analysis considers only *movement* dependencies and therefore will not apply to base-generated dependencies without additional assumptions. Second, because van Urk adopts the assumption that Agree is a precondition on movement (see e.g. Chomsky 2000, 2001b), he locates the A-/Ā-distinction in properties of features triggering Agree, whereas my proposal extends the relevant distinction among feature-types to Merge-triggering features. My proposal therefore straightforwardly accounts for the properties of base-generated dependencies constructed without Agree.

I will also note that van Urk's analysis does not provide a clear way to account for the Ban on Improper Movement, whereas mine does; see section §5.2.

Thus, my proposals are firmly embedded within a *feature-driven* approach to syntactic operations (on which see e.g. Adger 2003, Müller 2010, 2017, Abels 2012, Merchant 2014, Georgi 2017, Zyman 2018, 2024: esp. Supporting information, van Urk 2020, Hewett 2023, and Elbourne 2025), rather than a free or 'untriggered' approach (on which, see e.g. Chomsky 2013; 2015; 2020, Boeckx 2010, Ott 2010, 2015, Safir 2010, 2019, Epstein et al. 2014, 2015, Collins 2017, Chomsky et al. 2019, Milway 2019).

In the following two subsections, I posit two independently manipulable variables which interact to yield four feature and dependency types, all of which are attested in Iraqi. In section §4.1, I propose a distinction between two types of Merge-triggering features, expanding upon a proposal in Hewett (2023). Then, in section §4.2, I propose that the A-/Ā-distinction reduces to an independently necessary categorial distinction between DPs and QPs (the latter in the sense of Cable 2007, 2010; see also Safir's 2019 "HP"). Section §4.3 shows how four dependency types—A-movement, A-resumption, Ā-movement, and Ā-resumption—can be derived from these two hypotheses about features.

4.1 Two kinds of Merge-triggering features

I begin with the claim from Hewett (2023: esp. ch. 3.2) (inspired by ideas in McCloskey 2002 and work in the Minimalist Grammars framework, e.g. Stabler 1997, 2011 and Ermolaeva 2021) that external Merge and internal Merge are triggered by different types of features (*pace* e.g. Longenbaugh 2019: esp. 159, (31) and Newman 2021: esp. 24, (41)).[33] On Hewett's approach, external Merge is driven by '•' (read: 'bullet') features (notation from Heck and Müller 2007, *et seq.*), such as [•D•] which triggers external Merge of a DP. Internal Merge, by contrast, is driven by movement-specific features, for which Hewett proposes the diacritic '◁' (read: 'arrow'); thus, [◁D◁] will trigger internal Merge of a DP. In order to account for the properties of Iraqi A-resumption—in particular, those which differentiate A-resumption from A-movement—I will follow Hewett (2023) in distinguishing two types of Merge-triggering features.[34] But, whereas Hewett's analysis was exclusively focused on differentiating the triggers for external and internal Merge, I present an analysis which accommodates *parallel* Merge (e.g. Citko 2005).

Following Chomsky's (2001a) unificationist analysis of external and internal Merge as two subtypes of a single Merge operation, I propose that both •- and ◁-features trigger Merge but that they do so under different structural conditions. Concretely, I propose the definedness conditions on •- and ◁-triggered Merge in (52) and (53), respectively, both of which assume the definition of *matching* in (54) (modeled closely after a similar definition given in Zyman 2024: 21, (42)). I abbreviate the *root* of a syntactic object α as "R_α."

(52) **Definedness condition on •-triggered Merge**
Merge(α,β), where R_α or the head of which R_α is a projection bears [•F•] and is not irreflexively dominated and β matches [•F•], is defined iff there is no node γ such that R_α and R_β reflexively dominate γ.

(53) **Definedness condition on ◁-triggered Merge**
Merge(α,β), where R_α or the head of which R_α is a projection bears [◁F◁] and is not irreflexively dominated and β matches [◁F◁], is defined iff there is some node γ such that R_α and R_β reflexively dominate γ.[35]

[33] For a similar proposal made under quite different syntactic assumptions, see Elbourne (2025).

[34] I refer the reader to Hewett (2023: chs. 3–4) for empirical justification of this distinction among feature types.

[35] To account for minimality effects with internal Merge, we could add the following intervention condition to the definedness condition on ◁-triggered Merge in (53):

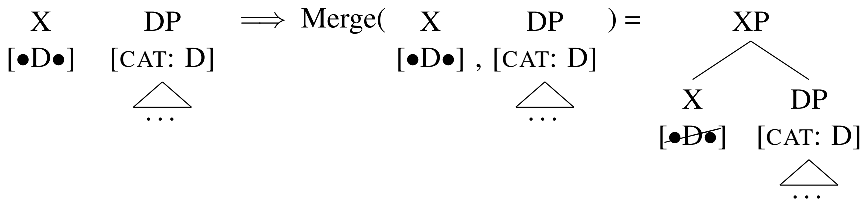
(i) ... and there is no distinct accessible syntactic object β' matching [◁F◁] and node γ' such that:
a. R_α and $R_{\beta'}$ reflexively dominate γ',
b. R_β does not dominate γ', and

(54) **Definition of *match***
For any Merge-triggering feature [{•F•, ◁F◁}] and syntactic object X with root R_X, X matches [{•F•, ◁F◁}] iff [{•F•, ◁F◁}] is keyed to the categorial feature of R_X.

In a nutshell, the difference between •-features and ◁-features is that •-features require the mergees to be completely separate (i.e. unconnected) syntactic objects, whereas ◁-features require the syntactic objects to be structurally connected via shared containment.

Let us consider some simple derivations to see how these features work in detail (for extensive discussion of the definition of Merge itself, see Zyman 2023a, 2024 and references therein). Example (55) considers a derivation with two distinct syntactic objects: a head X bearing the selectional feature [•D•] and a (potentially internally complex) DP bearing the categorial feature [CAT: D]. External Merge of X with DP, triggered by the [•D•] feature on R_X, is defined for X = α and DP = β, consonant with condition (52), because (i) R_X is not irreflexively dominated, (ii) R_X does not dominate any node reflexively dominated by R_{DP}, and (iii) DP matches [•D•] per (54).

(55) *External Merge is triggered by •-features*



Going forward, I will largely omit categorial features from the trees for simplicity.

Furthermore, note that the definedness condition on •-triggered Merge in (52) only requires that R_α (not R_β) be undominated prior to Merge. Consequently, if we adopt a version of the Extension Condition (see Chomsky 1995b: 190–191) along the lines of (56) (for a related definition, see Citko and Gračanin-Yuksek 2021: 20, (17)), •-features will not be limited to triggering external Merge.[36]

(56) *Extension Condition*
Merge(α,β) is defined iff at least one of {R_α, R_β} is not irreflexively dominated.

Rather, this formulation of the Extension Condition allows •-features to also trigger *parallel* Merge, where R_β is not dominated by R_α but is dominated by another node γ. Example (57) continues the derivation from (55): Y is a syntactic object whose root is undominated and bears a [•D•] feature, triggering Merge of Y with DP to form YP; this is parallel, not external, Merge because R_{DP} is irreflexively dominated prior to Merge. For explicitness, I represent the structures built by parallel Merge (and internal Merge, see below) as involving multidominance (see Citko 2005; Johnson 2009, 2012; and Citko and Gračanin-Yuksek 2021, as well as the many references cited in those works), though a copy theoretic approach involving sideward movement (e.g. Nunes 2004) could also be made to work.

(57) *Parallel Merge can be triggered by •-features*

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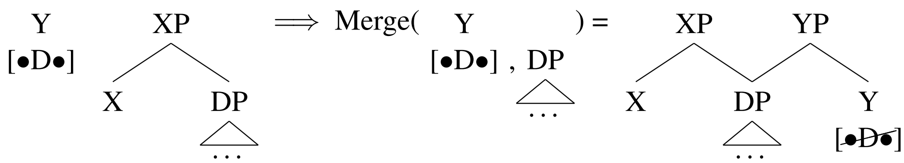
c. γ' c-commands γ.

Alternatively, see Hewett (2025c) for an explicit search algorithm for internal Merge that derives intervention effects and for arguments against a c-command based definition of 'closeness.'

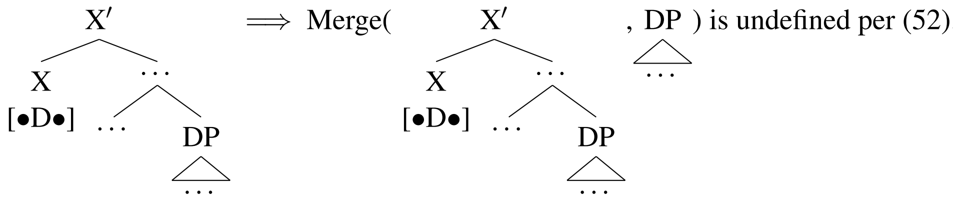
[36] In fact, the extension condition in (56) is redundant with the definedness conditions on Merge in (52)–(53), since the latter bake in the requirement for the head bearing the structure-building feature (i.e. the 'selector') to be undominated. For my analysis to be maintained, there must not be a more restrictive Extension Condition requiring all mergees to be undominated prior to Merge.



Below, I will argue that parallel Merge is crucially involved in creating the constituents which head Ā-dependencies, thereby accounting for many of the properties distinguishing Ā-dependencies from A-dependencies.

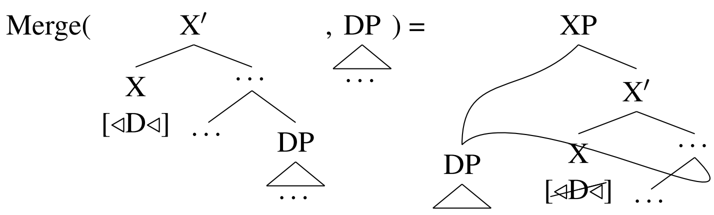
Let us turn now to consider derivations in which the roots of the syntactic objects α and β combined by Merge (reflexively) dominate one or more of the same nodes prior to Merge. As a shorthand in such cases, I will say that α and β 'share content.' For instance, consider the derivation in (58):[37] internal Merge of X′ with a DP that it dominates, triggered by a [•D•] feature on X,[38] will be undefined because $\mathrm{R_{X'}}$ and $\mathrm{R_{DP}}$ reflexively dominate $\mathrm{R_{DP}}$, in violation of (52).

(58) *Internal Merge is not triggered by •-features*



Crucially, however, ◁-triggered Merge in the same context *will* be defined per (53), forming XP, as illustrated below (again I represent multiple occurrences of a single syntactic object via multidominance):

(59) *Internal Merge is triggered by ◁-features*



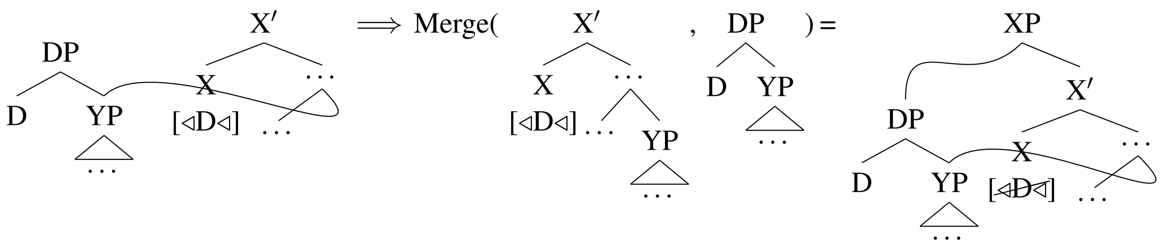
This is because there is at least one node (e.g. $\mathrm{R_{DP}}$ itself) which is reflexively dominated by both $\mathrm{R_{X'}}$ and $\mathrm{R_{DP}}$ prior to Merge—that is, X′ and DP share content.

Likewise, ◁-features can trigger (a form of parallel) Merge even when neither R_α nor R_β dominates the other, so long as α and β have shared content. Consider (60). Merge of X′ and DP triggered by an [◁D◁] feature on X is defined per (53) because there is at least one node (e.g. YP) which is mutually dominated by both $\mathrm{R_{X'}}$ and $\mathrm{R_{DP}}$. This is despite the fact that $\mathrm{R_{X'}}$ does not actually dominate $\mathrm{R_{DP}}$.

(60) *Parallel Merge where α and β have shared content is triggered by ◁-features*

[37] I use X′-notation only for convenience. All my trees could be made compliant with Bare Phrase Structure (Chomsky 1995a).

[38] See Adger (2003: 109–110), Merchant (2019: 326), and Zyman (2024: 19–28) on how a selectional feature lexically specified on a head can be satisfied by merging the selectee with a projection of the head, rather than with the head itself.



Derivations like (60)—where a constituent that was initially merged low in the clause (e.g. YP) gains an encapsulating 'shell' (e.g. DP) through parallel Merge, after which the shell is merged into the main clause—have been dubbed 'layering' derivations by Thoms (2019, 2023) and Thoms and Heycock (2022), and are also discussed by Johnson (2012), O'Brien (2017), Poole (2017), and Colley (n.d.) (and see Safir 2019 for a related proposal). In the present system, which adopts the definedness conditions in (52) and (53), such derivations can be generated without issue. This will prove vital to my analysis of Ā-movement presented in section §4.3 below.

To summarize, in this subsection, I have sketched an approach to structure-building which, following Hewett (2023), distinguishes two types of Merge-triggering features: •-features and ◁-features. I proposed two novel, explicit definedness conditions on Merge, each of which restricts the structural arrangements of mergees prior to Merge in a particular way.

4.2 A featural approach to the A-/Ā-distinction

In addition to differentiating two types of structure-building features, I adopt van Urk's (2015) proposal that the properties of A- and Ā-dependencies are attributable to syntactic features. While van Urk was exclusively concerned with features triggering Agree, I generalize this idea to cover both Merge and Agree features in the following conjecture:

(61) **A featural approach to the A-/Ā-distinction**
All differences between A- and Ā- dependencies derive from the features which construct those dependencies.

Because the A-/Ā-distinction is linked to syntactic features, no independent reference to (possibly stipulative) classes of A- and Ā-positions is necessary, contrary to traditional analyses.

To be truly satisfactory, the conjecture in (61) must be supplemented with an account of what makes a given feature an A-feature or an Ā-feature. van Urk (2015: 27–32) suggests that A-features are *obligatory* features of nominals, necessarily merged in the extended nominal projection (e.g. φ-features), whereas Ā-features related to *wh*-movement and the like are merely *optionally* present on syntactic objects. I will pursue a particular version of this idea, inspired by Safir's (2019) account of the A-/Ā-distinction, according to which the difference between A- and Ā-features is fundamentally related to *syntactic category*. Specifically, obligatory features of nominals are those associated with constituents of category 'D'; 'optional' features targeted by Ā-dependencies, on the other hand, are introduced by a head of a distinct syntactic category that is not part of the extended nominal projection, for which I adopt Cable's (2007; 2010) "Q" morpheme (see also Safir's 2019 "H"). In his analysis of pied-piping, Cable argues that the constituent which undergoes Ā-movement in a *wh*-question is not the *wh*-phrase itself (i.e. a DP bearing a [wh] feature), but is instead a QP which irreflexively dominates the *wh*-phrase. The Q morpheme that heads QP is overt in languages like Tlingit, Edo, and Sinhala, but is, by hypothesis, null in languages like Arabic and English. Following Cable (2010: ch. 6), I pursue a Q-based analysis for all types of Ā-dependencies, not just *wh*-movement.[39]

The core of my proposed categorial approach to the A-/Ā-distinction is summarized in (62).

[39] See Evile and Pesetsky (2023) for a related account of Ā-movement in relative clauses.

(62) **A categorial approach to the the A-/Ā-distinction**
A-features target (features of) DPs. Ā-features target (features of) QPs.

Because I am chiefly concerned with structure-building features, this hypothesis can be restated as follows:

(63) **A categorial approach to the A-/Ā-distinction, focused on structure-building features**
{[●D●], [◁D◁]} are A-features. {[●Q●], [◁Q◁]} are Ā-features.

On my analysis, what I have been referring to as 'A-features' and 'Ā-features' are, in fact, features which make reference to natural classes of constituents determined by their syntactic category—roughly, 'D-features' and 'Q-features,' respectively. Hence, my analysis reduces the A-/Ā-distinction to an independently motivated distinction among categorial features.

4.3 Four features, four dependency types

By combining the two preceding hypotheses about features—that there are two types of Merge-triggering features (section §4.1) and that the A-/Ā-distinction reduces to a distinction among categorial features (section §4.2)—we arrive at the four-way distinction among feature types in (64).

(64) *Typology of features predicted by two hypothesized distinctions among features*

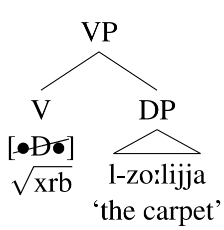
	A-feature	**Ā-feature**
◁-feature	[◁D◁] ↪ derives A-movement	[◁Q◁] ↪ derives Ā-movement
●-feature	[●D●] ↪ derives A-resumption	[●Q●] ↪ derives Ā-resumption

I will now argue that each feature can be linked to a different dependency type: [◁D◁] features trigger A-movement, [◁Q◁] features trigger Ā-movement, [●D●] features can trigger A-resumption, and [●Q●] features trigger Ā-resumption.

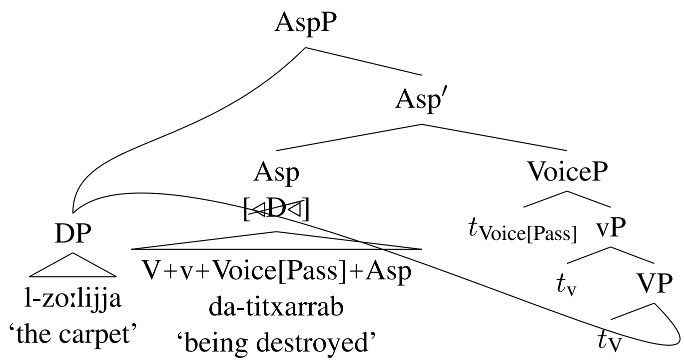
Consider first A-movement in the Iraqi passive sentence in (65).

(65) [$_{\text{TP}}$ l-zoːlijja$_k$ ʧaːnat da-titxarrab ___$_k$].
[$_{\text{TP}}$ the-carpet.FSG$_k$ AUX.PST.3FSG PROG-destroy.PASS.3FSG ___$_k$]
'The carpet$_k$ was being destroyed ___$_k$.' (Iraqi)

The derivation of this sentence proceeds as follows. First, a [●D●] feature on V triggers external Merge of V with the DP *l-zoːlijja* 'the carpet'; this complies with the definedness condition on ●-triggered Merge in (52) because V and DP do not have shared content prior to Merge.

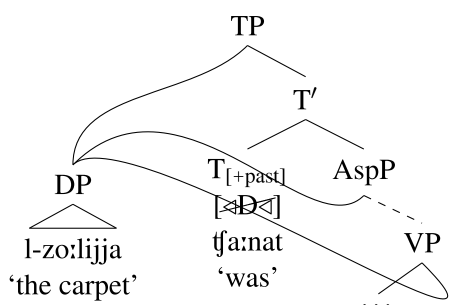
(66)


Then, the functional heads v, Voice[Pass], and Asp are merged into the structure, in that order (possibly triggered by ●-features, though I abstract away from this particular issue; see e.g. Pietraszko 2017, 2023a on building extended projections via c-selection). V raises cyclically to Asp and is realized there as *titxarrab* with the progressive aspect prefix *da-*; I defer accounting for main verb agreement until section §5.1. Importantly, in an A-movement derivation, I propose that Asp bears an [◁D◁] feature, which triggers internal Merge of DP into [Spec, AspP] in accordance with the definedness condition on ◁-triggered Merge in (53). Here and throughout, I only represent the structure-building features relevant to my analysis.

(67)


Since internal Merge of *l-zoːlijja* 'the carpet' is triggered by an [◁D◁] feature, it is A-movement per the hypothesis in (63).

Finally, T bearing an [◁D◁] feature is merged in and attracts the closest DP—in this case, the narrow subject—to its specifier (I use dashed branches to abbreviate irrelevant portions of the tree):[40]

(68)


Next, consider A-resumption. In a nutshell, I propose that resumptive A-dependencies in Iraqi are derived by (i) externally merging a pronoun in a low, thematic position, (ii) externally merging a DP in a higher, non-thematic position via a [•D•] feature, and (iii) binding the pronoun through lambda abstraction.[41] Thus, the key difference between A-resumption and A-movement lies in the featural composition of Asp. I will walk through a derivation along these lines for the Iraqi sentence in (69).

(69) [$_{\text{TP}}$ l-zoːlijja$_k$ ʧaːnat da-jindaːs *pro*$_{\text{EXPL}}$ ʕaleː-**ha**$_k$].
[$_{\text{TP}}$ the-carpet.FSG$_k$ AUX.PST.3FSG PROG-step.PASS.3MSG *pro*$_{\text{EXPL}}$ on-**3FSG**$_k$]
(lit.) 'The carpet$_k$ was being stepped on it$_k$.' (Iraqi)

[40]In declarative periphrastic tense clauses in Iraqi, the subject can either precede ((65)) or follow ((i)) a temporal auxiliary.

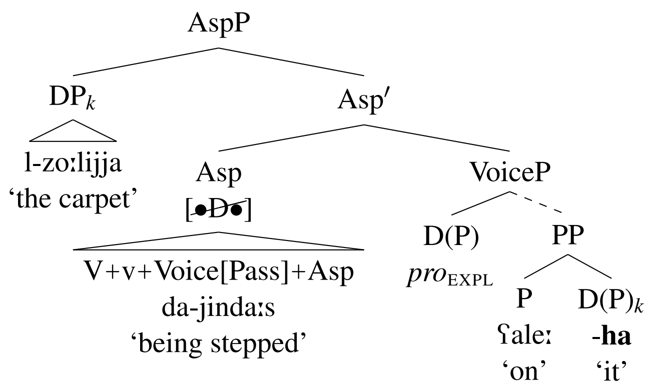
(i) [$_{\text{TP}}$ ʧaːnat l-zoːlijja$_k$ da-titxarrab ___$_k$].
[$_{\text{TP}}$ AUX.PST.3FSG the-carpet.FSG$_k$ PROG-destroy.PASS.3FSG ___$_k$]
'The carpet$_k$ was being destroyed ___$_k$.' (Iraqi)

I take this to reflect the optional presence of [◁D◁] features on clausal functional heads like T. See Cable (2012) and Zyman (2018) for related accounts of optional EPP-related subject movement in Dholuo and Janitzio P'urhepecha, respectively.

[41]Sellami (2024: esp. ch. 4) independently makes a very similar proposal.

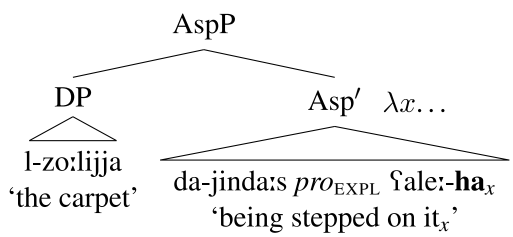
Building from the bottom up, the preposition *ʕala* 'on' externally merges with the pronoun *-ha* 'it.' The rest of the clause is then built up to VoiceP, including external Merge of the expletive narrow subject *pro*$_{\text{EXPL}}$ in [Spec, VoiceP] and head movement of the lexical verb.[42] The next head to be merged into the structure will be Asp. Crucially, I propose that the Iraqi lexicon contains an Asp bearing [•D•] instead of[43] [◁D◁]. This [•D•] feature is an A-feature per the hypothesis in (63) and it triggers *external* Merge of the broad subject DP *l-zoːlijja* 'the carpet' in [Spec, AspP], as shown in (70). Such Merge complies with the definedness condition on •-triggered Merge in (52) because the two mergees do not share content prior to Merge. Note, by contrast, that internal Merge of *pro*$_{\text{EXPL}}$ (or any other DP dominated by Asp′) with Asp′ to check the [•D•] feature on Asp is excluded by this same condition.[44]

(70)



In order to semantically integrate the broad subject into the rest of the clause, I propose that it binds a variable in Asp′ via lambda abstraction. In (69), this variable is the pronoun *-ha* 'it':[45]

(71)



Lastly, when T is merged in, its [◁D◁] feature will trigger internal Merge of the closest DP—in this case,

[42] I assume with Bruening (2013), Legate (2014), and Alexiadou et al. (2015), among others, that the implicit external argument of a passive sentence is not structurally represented, *pace* Collins (2005) and Merchant (2013), but nothing hinges on this.

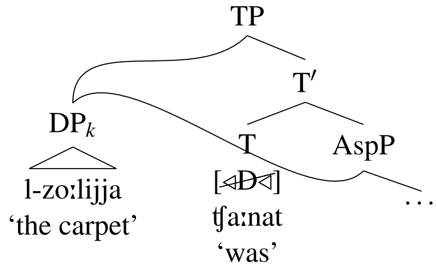
[43] Or perhaps in addition to, since overt narrow subjects can appear between AUX and the main verb, see (45). Because broad subjects always precede narrow subjects, if Asp can simultaneously bear [◁D◁] and [•D•], then these features must occur in an ordered list: Asp$_{[\langle ◁D◁,\ •D•\rangle]}$. See especially Müller (2010, 2014) and Georgi (2014, 2017) and references therein for other work positing ordered lists of Merge and Agree features on heads.

[44] If instead the Asp bearing just [◁D◁] were selected, *pro*$_{\text{EXPL}}$ would move to [Spec, AspP], be agreed with by T, and potentially also move to [Spec, TP], yielding the expletive pseudopassive sentence in (i) (see also (13)).

(i) [$_{\text{TP}}$ *pro*$_{\text{EXPL}}$ ʧaːn [$_{\text{AspP}}$ *pro*$_{\text{EXPL}}$ da-jindaːs [$_{\text{VoiceP}}$ *pro*$_{\text{EXPL}}$ ʕaleː-ha]]].
[$_{\text{TP}}$ *pro*$_{\text{EXPL}}$ AUX.PST.3MSG [$_{\text{AspP}}$ *pro*$_{\text{EXPL}}$ PROG-step.PASS.3MSG [$_{\text{VoiceP}}$ *pro*$_{\text{EXPL}}$ on-3FSG]]]
'It (f.) was being stepped on (lit. 'It$_{\text{EXPL}}$ was being stepped on it (f.).').'

[45] On an approach to binding like that of Büring (2004, 2005) where binding is achieved via syntactically represented binder prefixes, Predicate Abstraction is triggered by a μ-prefix (Büring 2004: 25, (2a–b)). Although Büring limits μ-prefixes to binding gaps created by movement (p. 25), Hewett (2023: ch. 7) crucially proposes that μ-prefixes are also responsible for binding resumptive pronouns in base-generated dependencies (see also Sellami 2024). Accordingly, Predicate Abstraction in (71) would be triggered by a μ-binder prefix adjoined immediately below the broad subject. For simplicity, I refrain from representing binder prefixes in the main text until section §5.5.

the broad subject—into its specifier. Example (72) illustrates.

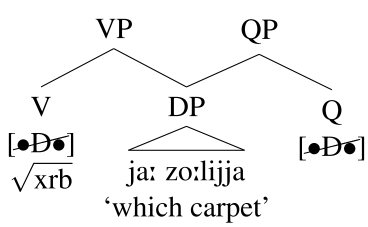
(72)


This completes the derivation of (69). Because this analysis of Iraqi A-resumption is novel (but see also Sellami 2024: ch. 4), I will reiterate its main ingredients here: (i) a (resumptive) pronoun is externally merged in a thematic position in the clause, (ii) the broad subject is externally merged in [Spec, AspP], triggered by a [•D•] feature lexically borne by Asp, and (iii) the broad subject binds the resumptive pronoun *qua* variable via lambda abstraction. Assuming that all languages utilize lambda abstraction and allow pronouns to be interpreted as variables, the key factor determining whether a language will have access to resumptive A-dependencies is the lexical distribution of [•D•] features: the lexicon of a language with A-resumption (e.g. Iraqi) contains functional heads like Asp which bear [•D•], whereas the lexicon of a language without A-resumption (e.g. English) lacks such a functional head.[46]

We can now turn to deriving Ā-movement, exemplified by the Iraqi *wh*-question in (73).

(73) [$_{\text{CP}}$ jaː zoːlijja$_k$ ʧaːn Matt da-jxarrub ___$_k$]?
[$_{\text{CP}}$ which carpet.FSG$_k$ AUX.PST.3MSG Matt PROG-destroy.3MSG ___$_k$]
'Which carpet$_k$ was Matt destroying ___$_k$?' (Iraqi)

Like Cable (2007, 2010), I analyze the heads of Ā-dependencies as QPs. However, unlike Cable, I do not take QPs to be initially generated in thematic positions, just like DPs. Instead, adapting ideas from Rezac (2003), Johnson (2012), O'Brien (2017), Poole (2017), Safir (2019), and Colley (n.d.), I propose that the QP shell is layered on to a DP late, after the DP has been externally merged in its thematic position. This layering is triggered by a [•D•] feature on Q.[47] Thus, in the derivation of (73), after the DP *jaː zoːlijja* 'which carpet' is externally merged with V (triggered by a [•D•] feature on V), I propose that Q's [•D•] feature causes the DP to undergo *parallel* Merge, forming QP:[48]

(74)


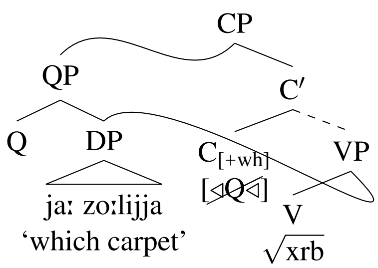
[46] In fact, things may not be so black and white. As McCloskey and Sells (1988: 178, fn. 11) and Rezac (2011: 274, fn. 22) note, English locative and experiencer *have* appears to select a base-generated non-thematic specifier that A-binds a resumptive pronoun, as in: [*These walls*]$_k$ *have a funny smell coming from* ***them***$_k$. Building on Myler (2014: sec. 4.1), we can analyze locative/experiencer *have* as a v bearing a [•D•] feature, which triggers external Merge of a DP in [Spec, v$_{have}$P]. This DP is then semantically integrated into the clause by binding a pronoun via lambda abstraction.

[47] In order to account for pied-piping of constituents larger than DP (e.g. PP), we will need to posit various lexical variants (or 'flavors') of Q which bear a selectional feature other than [•D•] (e.g. [•P•]).

[48] The exact timing of parallel Merge here is not important. It could also be the case that parallel Merge is delayed until more of the main clausal spine is built.

Recall that •-features are capable of triggering parallel Merge because the condition in (52) only requires the *selector* to be undominated prior to Merge.

Then, ignoring for the moment possible clause-internal intermediate landing sites, the derivation continues until it reaches C′. Updating Hewett's (2023) analysis of the trigger of *wh*-movement to the present system, I propose that $C_{[+wh]}$ is lexically specified to bear an [◁Q◁] feature. This feature will trigger (parallel) Merge of QP with C′, forming CP. This application of Merge is licensed by the condition on ◁-triggered Merge in (53) because R_{QP} and $R_{C'}$ mutually dominate at least one node (e.g. R_{DP}) prior to Merge.

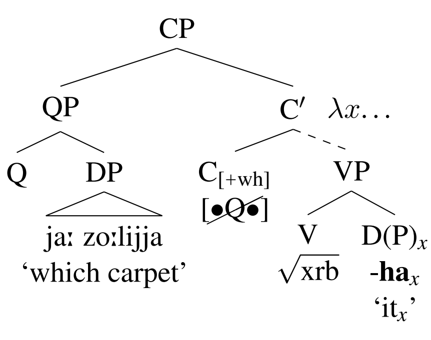
(75)


Furthermore, this instance of Merge can be characterized as Ā-movement per the hypothesis in (63) because it is triggered by an [◁Q◁] feature.

Finally, consider how the present system derives Ā-resumption in the following Iraqi *wh*-question:

(76) [$_{CP}$ jaː zoːlijja$_k$ ʧaːn Matt da-jxarrub-**ha**$_k$]?
[$_{CP}$ which carpet.FSG$_k$ AUX.PST.3MSG Matt PROG-destroy.3MSG-**3FSG**$_k$]
(lit.) 'Which carpet$_k$ was Matt destroying it$_k$?' (Iraqi)

Hewett (2023) argues that the heads of resumptive Ā-dependencies in Arabic are base-generated directly in [Spec, CP] where they bind a resumptive pronoun within their sister through lambda abstraction; the resumptive pronoun and its binder are not related via movement. Thus, building from the bottom of the tree, the pronoun *-ha* 'it' will be externally merged in the variable site as the complement to V. Abstracting over irrelevant details, the clause will be built up to TP. Then, a $C_{[+wh]}$ bearing a [•Q•] feature will be merged from the lexicon into the clause. This [•Q•] feature will trigger external Merge of the QP *jaː zoːlijja* 'which carpet'—which, being internally complex, I take to have been built up in a separate part of the workspace—in [Spec, CP], in compliance with the definedness condition on •-triggered Merge ((52)).

(77)


This completes my demonstration of how two hypothesized differences among structure-building features predict a four-way distinction among feature types that accounts for four attested dependency types. The table from (64) is repeated here for convenience.

(78) *Typology of features predicted by two hypothesized distinctions among features*

	A-feature	**Ā-feature**
◁-feature	[◁D◁] ↪ derives A-movement	[◁Q◁] ↪ derives Ā-movement
•-feature	[•D•] ↪ derives A-resumption	[•Q•] ↪ derives Ā-resumption

In the next section, I refocus the discussion on A-resumption and show how my analysis accounts for its mixture of A- and Ā-properties. Crucially, each of these properties is derivable from [•D•] features and their interaction with the rest of the grammar.

5 Deriving the mixed A-/Ā-properties of A-resumption from properties of structure-building features

An empirically adequate analysis of A-resumption must at a minimum account for its array of A- and Ā-properties summarized in the following table (repeated from (50)):

(79) *Comparison of the properties of A-movement, A-resumption, and Ā-movement in Iraqi* (completed)

	A-movement	**A-resumption**	**Ā-movement**
Can the head be agreed with by T?	✔	✔	✗
Can the head undergo A-movement?	✔	✔	✗
Can the head be controlled PRO?	✔	✔	–
Can the dependency license parasitic gaps?	✗	✗	✔
Does the dependency induce secondary WCO?	✗	✗	✔
Can the dependency bypass intervening DPs?	✗	✔	✔
Can the dependency span a finite clause?	✗	✔	✔

This section develops an analysis which derives each of these properties from aspects of structure-building features and their interaction with various independent grammatical constraints, including minimality and cyclicity. In brief, I argue (i) that the 'A'-properties of A-resumption stem from the binder being a DP rather than a QP and (ii) that the 'Ā'-properties of A-resumption, which distinguish A-resumption from A-movement, are only apparently Ā-related and are instead attributable to the lack of movement in resumptive A-dependencies.

5.1 T agreement is a property of DPs

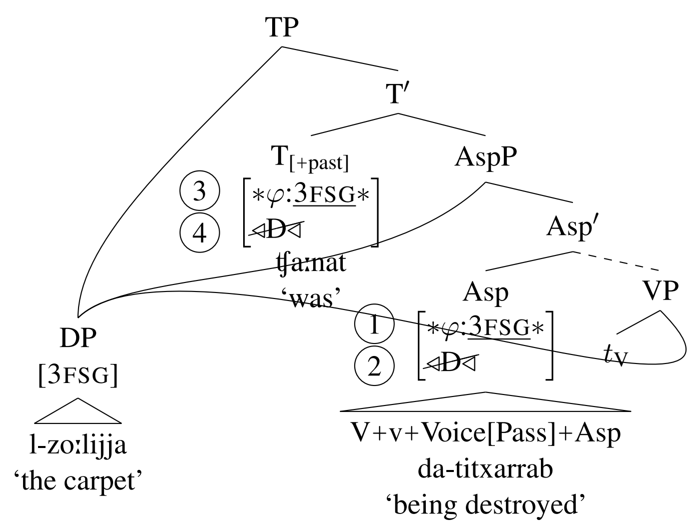
The first property to be derived is agreement with T, which is possible with the heads of A-movement and A-resumption dependencies but not with the head of an Ā-movement dependency. I follow Sellami (2024: ch. 7.4) and Kramer and Hewett (2024: sec. 4) and posit (at least) two distinct φ-probes in Arabic finite clauses: one on Asp and one on T (for a related proposal, see Soltan 2011: 246). In compound tenses, Asp's φ-probe is realized as agreement on the main verb (which has raised to Asp, see footnote 18), while T's φ-probe is realized as auxiliary agreement. Each φ-probe will initiate a search in the c-command domain of the head that bears it for the closest accessible goal with valued φ-features to agree with. I adopt the following standard, c-command–based definition of closeness:[49]

(80) Given three syntactic objects X, Y, and Z, Z is closer to X than Y is iff:
  a. X asymmetrically c-commands Z, and
  b. Z asymmetrically c-commands Y. (Hewett 2025c: 2, (3))

In a sentence with only a single accessible DP, such as (81), both φ-probes will agree with this DP.

(81) [$_{TP}$ l-zoːlijja$_k$ tʃaːnat da-titxarrab ___$_k$].
[$_{TP}$ the-carpet.FSG$_k$ AUX.PST.3FSG PROG-destroy.PASS.3FSG ___$_k$]
'The carpet$_k$ was being destroyed ___$_k$.' (Iraqi)

Because A-movement does not bleed agreement in Iraqi, I propose that [$*\varphi$:___$*$] features are ordered before [◁D◁] features when they cooccur on a head (see e.g. Georgi 2017 and Pietraszko 2023b on deriving interactions between agreement and movement through feature ordering). Example (82) illustrates how (81) is derived through interleaved agreement and A-movement. Circled numbers indicate the order in which features are checked during the derivation; features apply from top to bottom in a head's feature stack.

(82)


Asp agrees with the internal argument DP *l-zoːlijja* 'the carpet' (1) and internally merges it in [Spec, AspP] (2). Then, T agrees with DP in [Spec, AspP] (3) and internally merges it in [Spec, TP] (4).

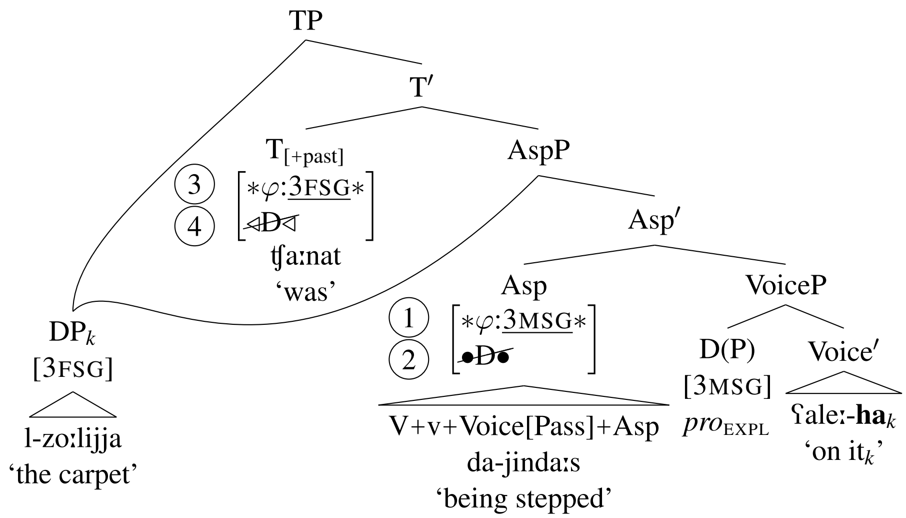
By contrast, in clauses with a broad subject, such as (83), the φ-probes on Asp and T find different goals.

(83) [$_{TP}$ l-zoːlijja$_k$ tʃaːnat da-jindaːs *pro*$_{EXPL}$ ʕaleː-**ha**$_k$].
[$_{TP}$ the-carpet.FSG$_k$ AUX.PST.3FSG PROG-step.PASS.3MSG *pro*$_{EXPL}$ on-**3FSG**$_k$]
(lit.) 'The carpet$_k$ was being stepped on it$_k$.' (Iraqi)

This is because the closest accessible goal to each head is distinct: for Asp, it is the narrow subject *pro*$_{EXPL}$ in [Spec, Voice[Pass]P], whereas for T, it is the broad subject *l-zoːlijja* 'the carpet' in [Spec, AspP]. Example (84) sketches the derivation of (83).

[49] But see Hewett (2025c) for arguments that asymmetric c-command is the wrong metric for calculating closeness, at least for evaluating minimality in A-movement.

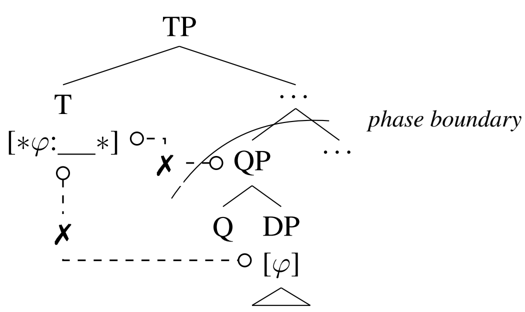
(84)



Asp agrees with pro_{EXPL} (1) and externally merges the broad subject DP *l-zoːlijja* 'the carpet' in [Spec, AspP] (2). Because the broad subject asymmetrically c-commands the narrow subject, it is the closest goal to T per the definition of closeness in (80) and, accordingly, T agrees with the broad subject (3) and internally merges it in [Spec, TP] (4). Thus, A-movement and A-resumption can feed agreement with T because both place a DP with valued φ-features close enough to T to be agreed with. The distribution of DPs in the clause and minimality constraints on Agree interact to derive which DP T agrees with.

To explain why the heads of Ā-dependencies in Iraqi cannot agree with T (see (17)), I contend that the shell which encapsulates Ā-moved *wh*-phrases is not a licit goal for φ-agreement and that it creates a barrier for agreement with the embedded *wh*-phrase (see Safir 2019: esp. sec. 3.1 for a related idea). Specifically, I propose that Q(P) lacks φ-features and that the complement of Q is opaque for φ-probing. Though there are various ways to account for this opacity, I will pursue a cyclicity-based account: Q is a phase head and its complement is rendered inaccessible at the point of probing by heads like Asp and T by being already spelled out (see the Phase Impenetrability Condition of Chomsky 2000: 108, (21)).[50] Consequently, even an intermediate step of Ā-movement of QP to a position above the narrow subject and below Asp or T (e.g. to [Spec, VoiceP]) will not feed φ-agreement, as illustrated schematically in (85).

(85)



To summarize, A-movement and A-resumption can feed φ-agreement with heads like T because the heads of A-dependencies are DPs with valued φ-features that are accessible to φ-probes. By contrast, because Ā-movement necessarily involves the addition of a QP shell over the moving *wh*-phrase, Ā-movement will never change which DP is the closest accessible φ-bearing goal to Asp or T.

[50] Alternatively, QPs could be *horizons* (in the sense of Keine 2019, 2020) for the φ-probes on Asp and T.

5.2 The ability to undergo A-movement is a property of DPs

The second property to be accounted for is the ability of a constituent to undergo A-movement (section §3.1.2): both narrow subjects ((86)) and broad subjects ((87)) can undergo A-movement, whereas heads of Ā-movement dependencies cannot ((88)), as per the Ban on Improper Movement (see (21)).

(86) [$_{\text{TP}}$ tˤilʕat [$_{\text{DP}}$ Joni]$_k$ [___$_k$ tʃuːf Matt]].
[$_{\text{TP}}$ turned.out.3FSG [$_{\text{DP}}$ Joni]$_k$ [___$_k$ hang.out.with.3FSG Matt]]
'Joni$_k$ turned out [___$_k$ to hang out with Matt].' (Iraqi)

(87) [$_{\text{TP}}$ tˤilʕat [$_{\text{DP}}$ Joni]$_k$ [___$_k$ da-jinðˤiħak *pro*$_{\text{EXPL}}$ ʕaleː-**ha**$_k$]].
[$_{\text{TP}}$ turned.out.3FSG [$_{\text{DP}}$ Joni]$_k$ [___$_k$ PROG-laugh.PASS.3MSG *pro*$_{\text{EXPL}}$ at-**3FSG**$_k$]]
(lit.) 'Joni$_k$ turned out [___$_k$ to be being made a fool of her$_k$].' (Iraqi)

(88) *[$_{\text{CP}}$ jaː bnajja$_k$ tˤilʕat [___$_k$ Matt jʃuːf ___$_k$]].
[$_{\text{CP}}$ which girl$_k$ turned.out.3FSG [___$_k$ Matt hang.out.with.3MSG ___$_k$]]
(int.) 'Which girl$_k$ turned out [Matt to hang out with ___$_k$]?' (Iraqi)

My analysis of this contrast is straightforward: A-movement is driven by [◁D◁] features, hence only DPs, and not QPs, can undergo A-movement. As I argued in sections §4.2–4.3, the heads of A-movement and A-resumption dependencies are DPs, merged into the structure via [◁D◁] and [•D•] features, respectively. As a result, DPs heading A-dependencies will be able to be undergo A-movement. By contrast, the heads of Ā-dependencies are necessarily QPs, merged into the clausal spine via [◁Q◁]/[•Q•] features, and so they cannot undergo Merge triggered by an [◁D◁] feature. Additionally, because Q's complement is inaccessible to syntactic operations outside QP due to QP being a phase (see section §5.1), A-movement of the DP complement of Q, stranding Q, is ruled out.[51]

5.3 The ability to be controlled PRO is a property of DPs

The next explanandum is the fact that narrow and broad subjects alike can be controlled PRO (section §3.1.3). The following examples are repeated from (26) and (29) above.

(89) Misk$_i$ gaːlat [$_{\text{CP}}$ inn-i nasˤsˤaħt Joni$_k$ [PRO$_{*i/k}$ tħasˤsˤil wðˤiːfa ʤdiːda]].
Misk.F$_i$ said.3FSG [$_{\text{CP}}$ that-1SG advised.1SG Joni.F$_k$ [PRO$_{*i/k}$ get.3FSG job new]]
'Misk$_i$ said that I advised Joni$_k$ [PRO$_{*i/k}$ to get a new job].' (Iraqi)

(90) Misk$_i$ gaːlat [$_{\text{CP}}$ inn-i nasˤsˤaħt Joni$_k$ [PRO$_{*i/k}$ jinqubaðˤ *pro*$_{\text{EXPL}}$ ʕaleː-**ha**$_{*i/k}$
Misk.F$_i$ said.3FSG [$_{\text{CP}}$ that-1SG advised.1SG Joni.F$_k$ [PRO$_{*i/k}$ arrest.PASS.3MSG *pro*$_{\text{EXPL}}$ on-**3FSG**$_{*i/k}$
min l-ʃurtˤa]].
by the-police]]
'Misk$_i$ said that I advised Joni$_k$ [PRO$_{*i/k}$ to be arrested (on) her$_{*i/k}$ by the police].' (Iraqi)

There are at least two possible accounts of this parallelism, both of which at least partly link a subject's ability to be controlled to its bearing a [CAT: D] feature.

On a PRO-based theory, the controlled subject is the pronominal element PRO (perhaps a *minimal pronoun* in the sense of Kratzer 2009, see Landau 2015: sec. 3.2).[52] If pronouns realize D^0 heads with silent NP complements (see e.g. Postal 1966, Elbourne 2001, Hewett 2023, and Arregi and Hewett 2024), then PRO too, as a pronoun, will bear a [CAT: D] feature. Consequently, PRO is expected to have the

[51] In order to block movement of DP through the edge of QP, I assume that Q lacks an [◁D◁] feature, though such a derivation might be independently ruled out as a violation of an Antilocality constraint, on which see Abels (2003: 12, (4)).

[52] For arguments in favor of PRO being pronominal, see Landau (2024a,b).

distribution of other (subject) DPs and to be externally merged into the structure with a [•D•] feature. PRO can be externally merged as an external argument as in (89), satisfying a [•D•] feature on v, or PRO can be externally merged as a broad subject as in (90), satisfying a [•D•] feature on Asp. In other words, because PRO is of category D, its distribution will track that of other DPs.

Alternatively, on a movement theory of control (e.g. Hornstein 1999, 2001; Boeckx et al. 2010), null subjects in control clauses are actually unpronounced lower copies of the A-moved controller DP. On this type of approach, the ability for both narrow and broad subjects to be controlled reduces to the ability of both types of subjects to undergo A-movement. I showed in section §5.2 that this parallelism follows if A-movement is driven by [◁D◁] features, as both narrow subjects and broad subjects are DPs.

In summary, on either a PRO- or movement-based theory of control, the parallelism between broad and narrow subjects vis-à-vis control is accounted for by the hypothesis that the controllee must be a DP.

5.4 Parasitic gap licensing is a property of QPs

The next asymmetry to be accounted for concerns parasitic gap licensing (section §3.1.4): A-movement ((91)) and A-resumption ((92)) do not license parasitic gaps, but Ā-movement does ((93)).

(91) * [$_{TP}$ [$_{DP}$ haðoːli l-mumaθθilaːt]$_k$ twaðˤðˤafu [$_{vP}$ t_V ___$_k$] [biduːn ma aqaːbil pg_k]].
[$_{TP}$ [$_{DP}$ these the-actresses]$_k$ hired.PASS.3PL [$_{vP}$ t_V ___$_k$] [without C interview.1SG pg_k]]
(int.) 'These actresses$_k$ were hired ___$_k$ [without me interviewing pg_k].' (Iraqi)

(92) * [$_{TP}$ [$_{DP}$ haðoːli l-banaːt] ʧaːnaw mbaːrak-**lhum**$_{\underline{DP}}$ [gabl ma nʃuːf $pg_{\underline{QP}}$]].
[$_{TP}$ [$_{DP}$ these the-girls] were.3PL congratulated-**3PL.DAT**$_{\underline{DP}}$ [before C see.1PL $pg_{\underline{QP}}$]]
(int.) 'These girls$_k$ were congratulated (lit. 'congratulated to them$_k$') before we saw pg_k.' (Iraqi)

(93) ? [$_{CP}$ jaː mumaθθilaːt$_k$ waðˤðˤafti [$_{vP}$ t_V ___$_k$] [biduːn ma tqaːbiliːn pg_k]]?
[$_{CP}$ which actresses.FPL$_k$ hired.2FSG [$_{vP}$ t_V ___$_k$] [without C interview.2FSG pg_k]]
'Which actresses$_k$ did you hire ___$_k$ [without interviewing pg_k]?' (Iraqi; Hewett 2023: 62, (78b))

While multiple possible analyses of the A-/Ā-distinction in parasitic gap licensing are conceivable, I will adopt a version of van Urk's (2015; 2017) analysis, as it is easily adapted to explain the parallelism between A-movement and A-resumption.[53] van Urk proposes that A-movement and Ā-movement create different kinds of predicates at LF: A-movement triggers abstraction over individuals (abbreviated 'x'), whereas Ā-movement triggers abstraction over choice functions (abbreviated 'f'); on the latter, see Reinhart (1998), Sauerland (1998), Ruys (2000), Sternefeld (2001), and Cable (2010), among others. van Urk combines this idea with Nissenbaum's (2000) analysis of parasitic gaps, according to which parasitic gap–containing constituents are derived predicates that are created by Ā-movement of a null operator to their edge[54] and which must be sister to a derived predicate of the same type at LF in order to be interpretable (e.g. via Predicate Modification). A predicate over individuals created by A-movement therefore will not be conjoinable with a predicate over choice functions created by Ā-movement of a null operator due to a type mismatch, correctly excluding parasitic gap licensing under A-movement.

We can straightforwardly extend van Urk's analysis to cover A-resumption in Iraqi by biuniquely linking the category of the head of the dependency to the type of variable that it binds. Suppose that DPs bind individual variables but QPs bind choice function variables (for more on interpreting QP movement via

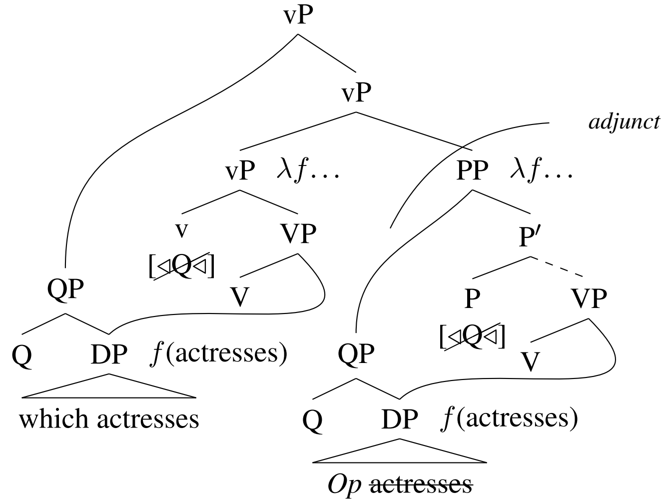
[53] See Hewett (2025b: Appendix A) for an alternative analysis which does not rely on different kinds of variable binding.

[54] For early proponents of the null operator movement analysis of parasitic gaps, see Contreras (1984, 1987), Stowell (1985: 491), Chomsky (1986: 54–68), and Browning (1987: ch. 3). See Hewett (2025b) for an analysis which dispenses with null operators in parasitic gap dependencies but which is nonetheless compatible with the proposals of the present paper.

abstraction over choice functions, see Poole 2017: esp. sec. 3.3.4).[55] Additionally, suppose that parasitic gaps are created by Ā-movement of a QP which contains a DP headed by a null operator *Op* whose NP complement is identical to that of the overt antecedent and is deleted under identity with it (see van Urk 2017: 540 for further discussion). Ā-movement of this QP to the edge of the adjunct island will create a derived choice function predicate (assuming that the null operator QP is not interpreted, or is interpreted as an identity function, at the edge of the island). This adjunct must then attach to a derived choice function predicate in the main clause to be interpreted with it via Predicate Modification.

As (94) illustrates for (93), Ā-movement of the QP 'which actresses' to an intermediate landing site—which I analyze as [Spec, vP] for explicitness—produces the licensing gap and creates the requisite derived choice function predicate. Note that, following Nissenbaum (2000), the parasitic gap–containing adjunct must attach *between* the moved phrase and its associated lambda abstract for compositional reasons (see Arregi and Murphy 2022: sec. 3.2 for additional relevant discussion). I refrain from representing the main clause external argument for simplicity in the following trees.

(94)

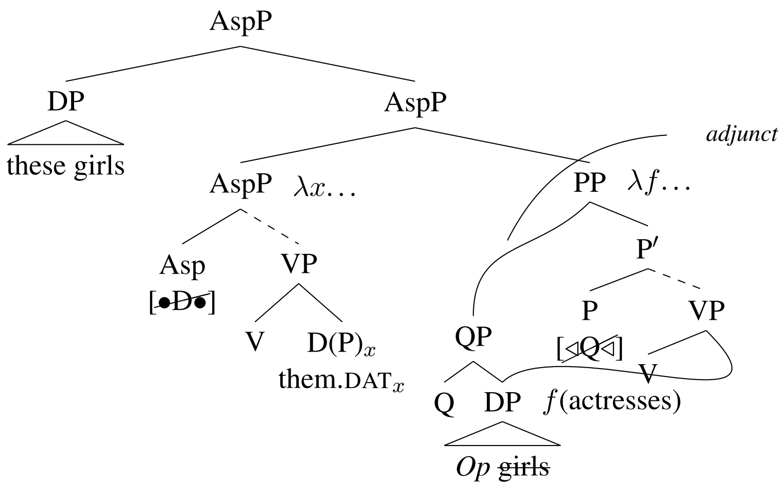


A-movement of a DP in the main clause, however, will create a derived predicate over *individuals*,[56] which cannot compose semantically with the adjunct *qua* predicate over choice functions due to a type mismatch. Parasitic gap licensing under A-resumption is ruled out for the same reason. Broad subjects are DPs, merged in a high non-thematic position via a [•D•] feature. Being DPs, broad subjects can only bind individual variables. The structure resulting from adjunction of a choice function predicate containing a parasitic gap to a derived predicate created by A-resumption, as in (95), will be uninterpretable.

[55]More needs to be said on this approach to explain how pronouns are interpreted. Because resumptive pronouns can terminate both A- and Ā-dependencies, they would need to be interpretable either as individual variables or choice function variables. However, van Urk's analysis of (primary weak) crossover relies on pronouns *only* having individual-denoting meanings. Thus, in order to be truly successful, a choice function analysis of parasitic gap licensing and crossover effects would need to posit some way to syntactically or semantically differentiate resumptive from non-resumptive pronouns for the purposes of binding. But note that doing so may run counter to the robust cross-linguistic generalization that resumptive pronouns are always morphologically identical to regular pronominal elements in the language in question (see Hewett 2023: ch. 7.5 for references and discussion).

[56]For a detailed exposition of how to interpret (A-)movement of a DP in a multidominant syntax, see Poole (2017: sec. 3.3.5).

(95) *



In summary, I have argued that parasitic gap licensing is only expected of Ā-movement dependencies[57] because, building on a proposal by van Urk (2015, 2017), Ā-dependencies create a different kind of derived predicate than A-dependencies do. I proposed a biunique relationship between the category of the head of the dependency (i.e. the binder) and the type of variable that it binds: DPs bind individual variables, while QPs bind choice function variables. The next section will show that another asymmetry between A- and Ā-dependencies can be linked to an interaction between grammatical category and variable binding.

5.5 The ability to indirectly bind is a property of DPs

A-dependencies, including A-resumption ((96)), do not trigger secondary weak crossover effects. Ā-dependencies do ((97)).

(96) [$_{TP}$ [$_{DP}$ ʤantˁat kull bnajja$_i$]$_k$ ʧaːnat [ism-ha$_i$] da-jinkitab ʕaleː-**ha**$_k$].
[$_{TP}$ [$_{DP}$ bag.FSG every girl$_i$]$_k$ AUX.PST.3FSG [name.MSG-her$_i$] PROG-write.PASS.3MSG on-**3FSG**$_k$]
'[Every girl$_i$'s bag]$_k$ had [her$_i$ name] being written on it$_k$.' (Iraqi)

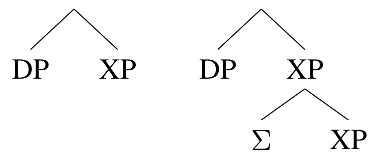
(97) *[$_{CP}$ [ʕala ʤantˁat jaː bnajja$_i$]$_k$ ʧaːn [ism-ha$_i$] da-jinkitab ___$_k$]?
[$_{CP}$ [on bag.FSG which girl$_i$]$_k$ AUX.PST.3MSG [name.MSG-her$_i$] PROG-write.PASS.3MSG ___$_k$]?
(int.) '[On which girl$_i$'s bag]$_k$ was [her$_i$ name] being written ___$_k$?' (Iraqi)

As discussed in section §3.1.5, I follow Büring (2004, 2005) and Hewett (2023, 2025a) in analyzing secondary crossover effects as the failure of a quantifier contained within a larger constituent XP to bind a(n E-type) pronoun outside of XP. I refer to such binding out as 'indirect binding.' In particular, the empirical contrast in (96)–(97) indicates that indirect binding is possible out of DPs but not out of QPs. I will lay out two possible ways to derive this difference, one syntactic and one semantic, both crucially linked to the categorial difference between DP and QP, in line with my overall approach to the A-/Ā-distinction. I must, however, leave for future work the task of deciding between these (and other) explanations.

The first possibility is to syntactically restrict where indirect binding can take place. Büring (2004, 2005) develops an analysis of binding in which quantifiers do not bind anything. Rather, binding is mediated by structurally represented binder prefixes which are adjoined immediately below quantifiers. In his system, the binder prefix responsible for indirect binding ('Σ') is excluded from Ā-positions (see Hewett 2023: 398–400). We can adapt this proposal to the present system and eschew any reference to A- and Ā-positions by restricting Σ to environments with a DP antecedent:

[57] See Hewett (2023: secs. 3.4–3.5) on why base-generated resumptive Ā-dependencies in Arabic do not license parasitic gaps in clause-mate, clause-final adjuncts.

(98) Σ-*adjunction rule*
(optionally)



Because the structural description of this rule includes a DP, we account for the fact that indirect binding out of DPs, including broad subjects, is possible ((96)). Crucially, this rule will never adjoin a Σ-binder below a QP, explaining why Ā-dependencies induce secondary weak crossover effects ((97)).

A second possibility would be to allow Σ-adjunction (and therefore indirect binding) to apply freely, *pace* Büring (2004) and Hewett (2023), and to filter out semantically deviant derivations at the interfaces. Büring (2004: 45–47) argues that indirect binding involves abstraction over an individual variable and a situation variable (see e.g. Kratzer 1989, Heim 1990, and Elbourne 2005, 2013 on situation semantics). Recall from section §5.4 that I have proposed that DPs bind individual variables while QPs bind choice function variables, adapting ideas from van Urk (2015, 2017). Consequently, DPs are expected to be compatible with a Σ-binder. QPs, on the other hand, will be unable to semantically compose with the predicate over individuals and situations created by Σ, since QPs necessarily bind choice functions.

In summary, a categorial approach to the A-/Ā-distinction makes available at least two ways to account for the distribution of secondary weak crossover effects, neither of which needs reference to A- or Ā-positions. Instead, contrasts such as (96)–(97) emerge from interactions between constraints on binding and syntactic categories.

5.6 Intervention and clause-boundedness are properties of ◁-features

The final two properties to be accounted for are intervention effects and clause-boundedness. As I showed in section §3.2, A-resumption behaves like Ā-movement and unlike (at least some types of) A-movement in skipping intervening (non-*wh*) DPs ((99)) and spanning finite clause boundaries ((100)):

(99) a. * [$_{TP}$ [$_{DP}$ l-fiːzaːt]$_k$ inminħat [$_{DP}$ l-tˤullaːb] ___$_k$].
[$_{TP}$ [$_{DP}$ the-visas.FPL]$_k$ awarded.PASS.3FSG [$_{DP}$ the-students.MPL] ___$_k$]
(int.) 'The visas$_k$ were awarded the students ___$_k$.' (Iraqi)

b. [$_{CP}$ jaː fiːzaːt$_k$ inminħaw [$_{DP}$ l-tˤullaːb] ___$_k$]?
[$_{CP}$ which visas.FPL$_k$ awarded.PASS.3PL [$_{DP}$ the-students.MPL] ___$_k$]
'Which visas$_k$ were the students awarded ___$_k$?' (Iraqi)

c. [$_{TP}$ [$_{DP}$ haðoːli l-bnaːt]$_k$ ʧaːnaw [$_{DP}$ hwaːja iʃaːʕaːt] da-tinkitab ʕan-**hum**$_k$ b-l-ʤariːda].
[$_{TP}$ [$_{DP}$ these the-girls]$_k$ AUX.PST.3PL [$_{DP}$ many rumors.FPL] PROG-write.PASS.3FSG about-**them**$_k$ in-the-newspaper]
'These girls$_k$ had many rumors name being written about them$_k$ in the newspaper.' (Iraqi)

(100) a. * [$_{TP}$ [$_{DP}$ haːj l-sajjaːra]$_k$ ʧaːnat maʃkuːka [$_{CP}$ innu bigti ___$_k$]].
[$_{TP}$ [$_{DP}$ this.FSG the-car.F]$_k$ AUX.PST.3FSG suspected.FSG [$_{CP}$ that stole.2FSG ___$_k$]]
(int.) 'This car$_k$ was suspected that you stole ___$_k$ (∼ to have been stolen by you).' (Iraqi)

b. [$_{CP}$ jaː sajjaːra$_k$ ʧaːn *pro*$_{EXPL}$ maʃkuːk [$_{CP}$ innu bigti ___$_k$]].
[$_{TP}$ which car.F$_k$ AUX.PST.3MSG *pro*$_{EXPL}$ suspected.MSG [$_{CP}$ that stole.2FSG ___$_k$]]
'Which car$_k$ was it suspected that you stole ___$_k$?' (Iraqi)

c. [$_{\text{TP}}$ [$_{\text{DP}}$ haːj l-sajjaːra]$_k$ ʧaːnat *pro*$_{\text{EXPL}}$ maʃkuːk [$_{\text{CP}}$ innu bigtiː-**ha**$_k$]].
[$_{\text{TP}}$ [$_{\text{DP}}$ this.FSG the-car.F]$_k$ AUX.PST.3FSG *pro*$_{\text{EXPL}}$ suspected.MSG [$_{\text{CP}}$ that stole.2FSG-**3FSG**$_k$]]
(lit.) 'This car$_k$ was suspected that you stole it$_k$ ($\sim$ to have been stolen by you).' (Iraqi)

In previous sections, I analyzed the parallels between A-movement and A-resumption as natural class behavior, since both types of dependencies are built with 'D'-features. It might seem natural, therefore, to seek a featural natural class for A-resumption and Ā-movement to explain the facts in (99)–(100). Yet, my feature system does not make such a natural class available: the [•D•] features triggering A-resumption and the [◁Q◁] features triggering Ā-movement have no shared properties.

Instead, I contend that DP-intervention and clause-boundedness are characteristic of the [◁D◁] features which trigger A-movement. A-resumption and Ā-movement only appear to pattern alike because neither is generated with [◁D◁] features—this is an elsewhere pattern. To begin, I propose that minimality and locality constraints apply exclusively to features which initiate a search through a tree. Internal Merge-triggering ◁-features and Agree–triggering φ-probes are two such features. •-features, which trigger external or parallel Merge with a syntactic object unrelated to the selector, are not.

Now, consider why A-movement displays DP-intervention: an [◁D◁] feature will initiate a search in the sister of its bearer for the closest accessible matching syntactic object (see footnote 35). Because the [◁D◁] feature is keyed to constituents of category D (see the definition of *match* in (54)), we predict DP-relativized minimality, as in (99a). Neither [◁Q◁] features nor [•D•] features initiates a search through the tree for a constituent bearing [CAT: D], correctly predicting a lack of DP-intervention in Ā-movement ((99b)) and A-resumption ((99c)), respectively.

Next, consider why A-movement displays clause-boundedness, as in (100a). There are at least two reasons why finite CPs might be opaque to A-movement.

First, it could be that CPs—or more precisely, complements of C—are opaque by virtue of C being a phase head. Per the Phase Impenetrability Condition (Chomsky 2000: 108, (21)), in order for a constituent to escape C's complement, it must first move to CP's edge. We can explain why A-movement cannot escape finite CPs in many languages by lexically restricting the distribution of [◁D◁] features. Specifically, in languages like English and Iraqi, finite C does not bear [◁D◁]. Without an [◁D◁] feature, finite C will not attract a DP to its specifier, and we correctly exclude A-movement out of finite CPs.[58]

A second possibility is that CPs are *selectively opaque* to [◁D◁] features, in the sense of Keine (2019, 2020). Based on the observation that some maximal projections only block certain kinds of movement/agreement across them, Keine argues that a feature's search space can be idiosyncratically fixed by a node he calls a *horizon*. Horizons are defined in terms of categorial features and halt a probe's search procedure once a constituent of the relevant category is reached. On this approach, A-movement can be prevented from escaping CP if all [◁D◁] features in the matrix clause have [CAT: C] as a horizon. This systematic behavior across heads bearing [◁D◁] features could be viewed as the result of a *default* horizon setting, on which see Keine (2019: 54–55; 2020: sec. 3.7).[59]

Crucially, whichever analysis of the clause-boundedness of A-movement we adopt, it will correctly not extend to A-resumption in (100c), since the latter is not triggered by a ◁-feature. This is because Merge triggered by a [•D•] feature is insensitive to the internal structure of the selector, including whether or not it is or contains a finite CP. Furthermore, we can explain the non-clause-boundedness of Ā-movement in

[58] On this approach, languages with hyperraising would be characterized by having a (finite) C bearing [◁D◁]; see Zyman (2023b: sec. 4) for relevant discussion and references.

[59] A third possible account of CP opacity, following Halpert (2019), would be to invoke (relativized) A-over-A locality. This would be similar to the horizons account, but it would crucially derive CP opacity from shared properties of DPs and CPs. Suppose that finite CPs also match the trigger for A-movement (which would need to be something other than [◁D◁]; see Hewett 2025c: sec. 4.1). Then, a CP α will intervene for A-movement of any constituent (e.g. DP, CP) that α irreflexively dominates. Though initially promising, see Zyman (2023b: sec. 3.2) for some challenges facing an A-over-A intervention analysis of the lack of hyperraising in English.

(100b) in one of two ways. On a phasal approach to CP opacity, C would bear an [◁Q◁] feature to trigger Ā-movement to CP's edge, feeding movement of QP out of CP. On a horizons approach to CP opacity, no head bearing [◁Q◁] would have [CAT: C] as a horizon, permitting QP movement out of CP.

To summarize, only features which initiate a search through a tree, such as ◁-features and φ-probes, interact with minimality and locality constraints. The relativized minimality and locality profile of A-movement is derivable from properties of the [◁D◁] feature that triggers it. Because A-resumption involves base-generation, triggered by a [•D•] feature, no minimality or locality effects are expected. As (101) shows, a broad subject can bind a resumptive pronoun across an intervening CP and DP without issue.

BIND

(101) $[_{\text{TP}}$ T $[_{\text{AspP}}$ $\text{DP}_{\text{Broad Subj}}$ λx $\text{Asp}_{\cancel{[\bullet \text{D} \bullet]}}$ … $[_{\text{CP}}$ … DP … $\textbf{RP}_x$ …]]]

Not interveners for binding

These differences between A-resumption and A-movement are not trivial and, importantly, they militate against an alternative analysis according to which A-resumption involves run-of-the-mill A-movement but with an (exceptionally) overtly realized lowest copy of movement (see Hewett 2023: sec. 5.8.2 for references and critical discussion of spelled-out trace approaches to Ā-resumption). Such an alternative cannot straightforwardly explain the divergent behavior of A-movement and A-resumption vis-à-vis minimality and locality, so I reject it as both descriptively and explanatorily inadequate.

5.7 Summary

Let us take stock. I have argued that a feature-driven approach to structure building which posits a four-way distinction among feature types—[◁D◁], [•D•], [◁Q◁], and [•Q•][60]—is sufficient to account for the A-/Ā-distinction. In particular, I proposed that the relevant differences between A- and Ā-dependencies hinge on two factors: (i) the *category* of the XP heading the dependency (DP vs. QP) and (ii) whether the dependency is formed via movement or base-generation. No independent reference to classes of A- or Ā-positions is necessary, in line with van Urk (2015) and Safir (2019). Importantly, I showed that the 'mixed' A-/Ā-properties of Iraqi A-resumption documented in section §3 are derivable from two aspects of [•D•] features: the A-properties of A-resumption follow from [•D•] triggering merge of a DP and the 'Ā'-properties of A-resumption follow from the lack of movement in A-resumption.

6 Conclusion

I have argued that A-resumption in Iraqi provides a novel window into the underpinnings of the A-/Ā-distinction. A-resumption displays a number of the hallmark properties of A-dependencies despite not being generated by movement, demanding an analysis of the A-/Ā-distinction that is decoupled from the mechanics of movement (cf. e.g. van Urk 2015 and Safir 2019). What's more, I demonstrated that A-resumption diverges from A-movement in exhibiting a mix of A- and Ā-properties—a discovery which challenges traditional, position-based analyses of the A-/Ā-distinction which predict a strict bipartition among dependencies.

At its heart, my analysis of A-resumption and the A-/Ā-distinction rests on the following two claims: (i) there is a distinction between A- and Ā-features which reduces to the category of the constituent targeted by that feature (i.e. D vs. Q), and (ii) there is a distinction between the featural triggers for external Merge and internal Merge (i.e. • vs. ◁). Both claims have precedents in the literature, but this is the first time to my knowledge that they have been brought together. I hope to have shown that, by combining these hypotheses about features, we can account for the properties of Iraqi A-resumption—a success arguably not shared by

[60] It goes without saying that this does not exhaust the possible inventory of structure-building features.

conceivable alternatives (see especially footnote 32 and Appendix A). As a final note, I will point out that my analysis explains the mixture of A- and Ā-properties in A-resumption without relying on 'mixed' positions or probes (cf. van Urk 2015 *et seq.*). It remains to be seen whether apparent mixed A-/Ā-movement in other languages can be reanalyzed without recourse to conjunctive or composite probing; for instance, see Keine and Zeijlstra (2023) for a leapfrogging A-movement approach to apparent mixed A-/Ā-movement.

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A Comparison to Safir (2019)

Safir (2019) develops an account of the A-/Ā-distinction which, like my own analysis, is essentially *categorial* (though he does not use that term): the category of a constituent heading an A-dependency is distinct from the category of a constituent heading an Ā-dependency. On his analysis, this categorial difference results from countercyclically[61] merging an 'insulating' shell over an element undergoing Ā-movement. Safir dubs this head H, equivalent to my Q. Crucially, Safir maintains that Merge (including insulating Merge of an HP) applies freely, without being triggered by features, and is at most indirectly forced by interface conditions (see the references cited in Hewett 2023: ch. 4 for other work in the free Merge framework).

The indirect trigger for insulation can be seen by considering how *wh*-movement in (102) is derived.

(102) Who$_1$ did Mary praise ___$_1$?

Assume that *wh*-movement must pass through [Spec, vP] per the Phase Impenetrability Condition. Then, intermediate movement of the DP *who* in (102) to an outer specifier of v, asymmetrically c-commanding the external argument DP *Mary* in [Spec, vP], would normally prevent T from agreeing with and assigning nominative Case to *Mary* by Minimality considerations, because DP *who* would intervene. If all DPs must be assigned Case in order for the derivation to converge (i.e. the Case Filter), *wh*-movement of *who* should cause the derivation to crash. Safir argues, however, that this crash can be prevented by insulating the moving *wh*-object, rendering it a non-intervener for Agree and case assignment. Safir then argues roughly that A-properties are properties of DPs and that Ā-properties are properties of insulated DPs, namely HPs (which are non-DPs). For instance, binding from a derived position is possible for DPs (which head A-dependencies) but not HPs (which head Ā-dependencies), accounting for crossover effects, among others.

Although our analyses have much in common, a key difference lies in how each approaches syntactic structure building: on my analysis, Merge is feature-driven, whereas on Safir's analysis, Merge is free. There are empirical reasons, however, to favor my feature-driven approach. First, as Safir (2019: 312–313) discusses, the free insulation approach to the A-/Ā-distinction predicts that insulation should be unnecessary for *wh*-phrases base-generated in a high position, for instance in resumptive Ā-dependencies. A high base-generated operator will not interfere with case or agreement within the clause, so we predict that insulation should be merely optional. Safir argues that this is a welcome consequence, citing the following English example as evidence that primary weak crossover effects are absent in resumptive restrictive relative clauses:

(103) She was always going out with the kind of guy who$_1$, after you meet him$_1$, you have the feeling that he$_1$ is just like the last one she went out with. (slightly adapted from Safir 2019: 312, (46))

However, as I discussed in section §3.1.5, such examples suffer from a confounding ambiguity in identifying the resumptive variable and therefore do not provide sufficient evidence for the alleged lack of crossover under resumption. On the contrary, Hewett (2023: ch. 7, 2025a) has shown that, when the confounding ambiguity is eliminated, as in secondary crossover configurations, crossover effects are clearly detectable in base-generated Ā-dependencies, as in the following Iraqi Arabic resumptive *wh*-question:

[61] Safir (2019: esp. sec. 3) rejects the Extension Condition (Chomsky 1995b) and the Strict Cycle Condition (e.g. Chomsky 1973: 243) in favor of a weaker condition he calls the *Peak Novelty Condition* to allow for countercyclic Merge (see Müller 2023: 40–41 for discussion).

(104) [asˤdiqaːʔ minu$_i$]$_k$ ma ʧinti ħaːðˤira [lamman sˤaːħibt-a$_{*i/j}$ ʕarrafat-na
[friends who$_i$]$_k$ NEG were.2FSG present.FSG [when girlfriend.F-his$_{*i/j}$ introduced.3FSG-1PL
ʕaleː-**hum**$_k$]?
to-**them**$_k$]
(lit.) '[Whose$_i$ friends]$_k$ were you absent [when his$_{*i/j}$ girlfriend introduced us to them$_k$]?' (Iraqi)

Note that secondary weak crossover in (104) cannot be attributed to Ā-movement of the *wh*-phrase *asˤdiqaːʔ minu* 'whose friends' from the position of the resumptive pronoun *-hum* 'them', because the latter occurs in an adjunct island and islands are opaque to movement (see Hewett 2023: ch. 7.4.4, 2025a: sec. 4.3 for additional evidence of crossover persisting with in-island resumption).

On Safir's analysis, there is no clear motivation for insulating *asˤdiqaːʔ minu* 'whose friends' if it is base-generated in the matrix [Spec, CP]. Without insulation, we would expect (indirect) binding to be possible by the base-generated DP head of the dependency—parallel to what we see with the broad subject in (96)—contrary to fact. Safir's analysis thus fails to predict the presence of crossover effects in base-generated Ā-dependencies like (104) without additional assumptions. On my analysis, (104) is accounted for because $C_{[+wh]}$ in such sentences bears a [•Q•] feature, triggering external Merge of a QP in [Spec, CP]; layering of QP is forced by the featural requirements of $C_{[+wh]}$. Because indirect binding is impossible out of QP (see section §5.5), we derive the secondary weak crossover effect.

My feature-driven approach to Merge also offers a natural explanation for cross-linguistic variation in the distribution of base-generated resumptive dependencies. Consider A-dependencies: Iraqi employs A-resumption with broad subjects, while English seems not to (as indicated by the unacceptability of many of the translations of the Iraqi examples throughout this paper). My analysis locates this difference in the featural properties of lexical items: Iraqi has in its lexicon an Asp bearing [•D•], while English does not. On the free Merge/insulation approach of Safir (2019), such a difference would need to arise at the interfaces. Yet, I can think of no plausible interface-related difference that would derive this asymmetry. There is no obvious (morpho)phonological difference between the two languages that would predict the (non)availability of A-resumption at PF. Furthermore, there doesn't appear to be anything particularly special about the semantics of broad subjects that would explain at LF why they are present in Iraqi but absent from English: broad subjects combine with their sisters through run-of-the-mill Predicate Abstraction. I therefore conclude that a feature-driven approach to Merge provides a more straightforward account of the (non)availability of A-resumption across languages. See Hewett (2023: ch. 4) for an additional argument in favor of feature-driven Merge over free Merge from cross-linguistic variation in the availability of mixed Ā-chains.